

COMPLETENESS

PRECISION IN ABSTRACT NON INTERFERENCE

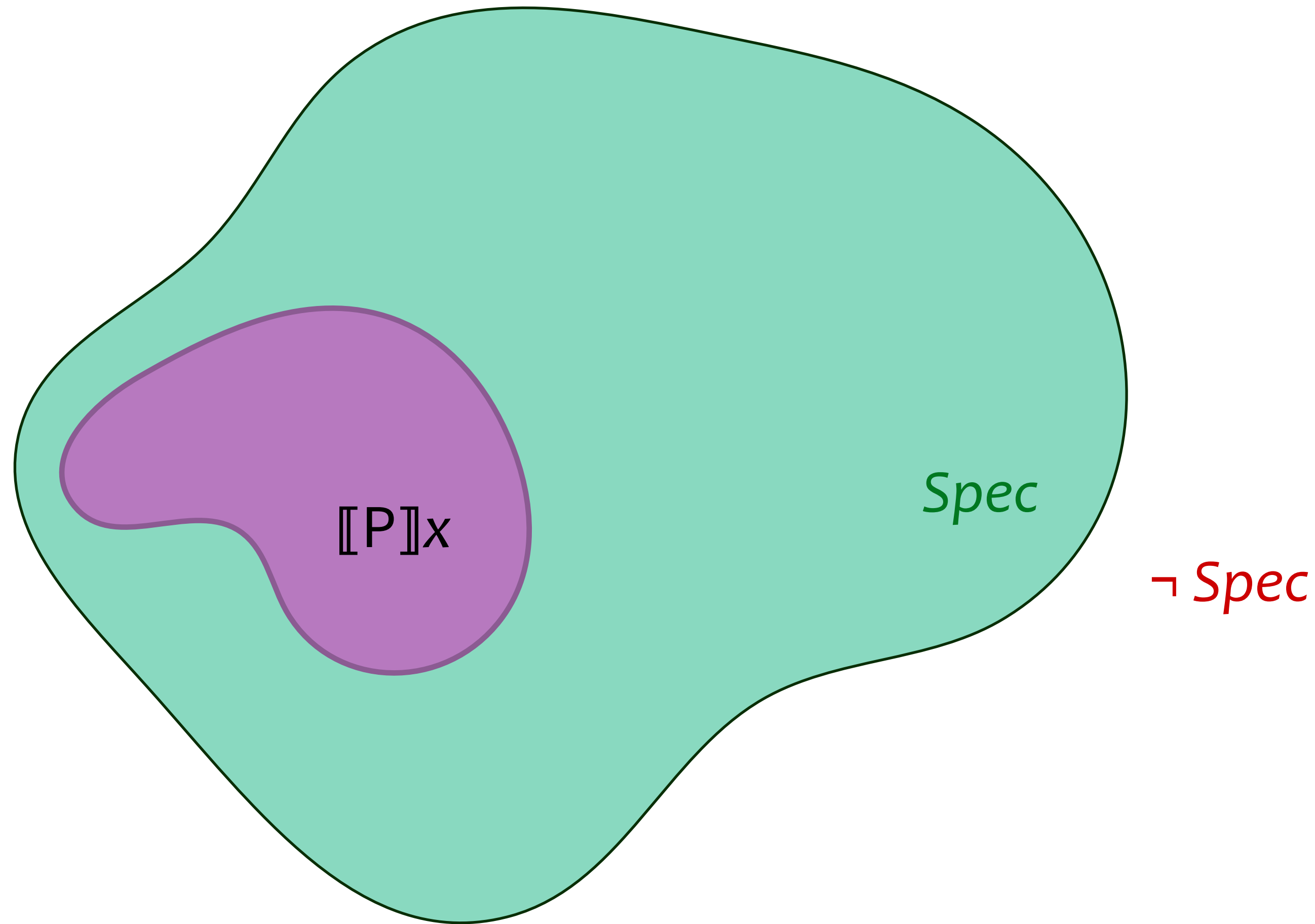
ISABELLA MASTROENI

PARMA - NOVEMBER 26TH 2025

VERIFICATION PROBLEM

➤ $\llbracket P \rrbracket x \subseteq \text{Spec}$?

Not decidable!



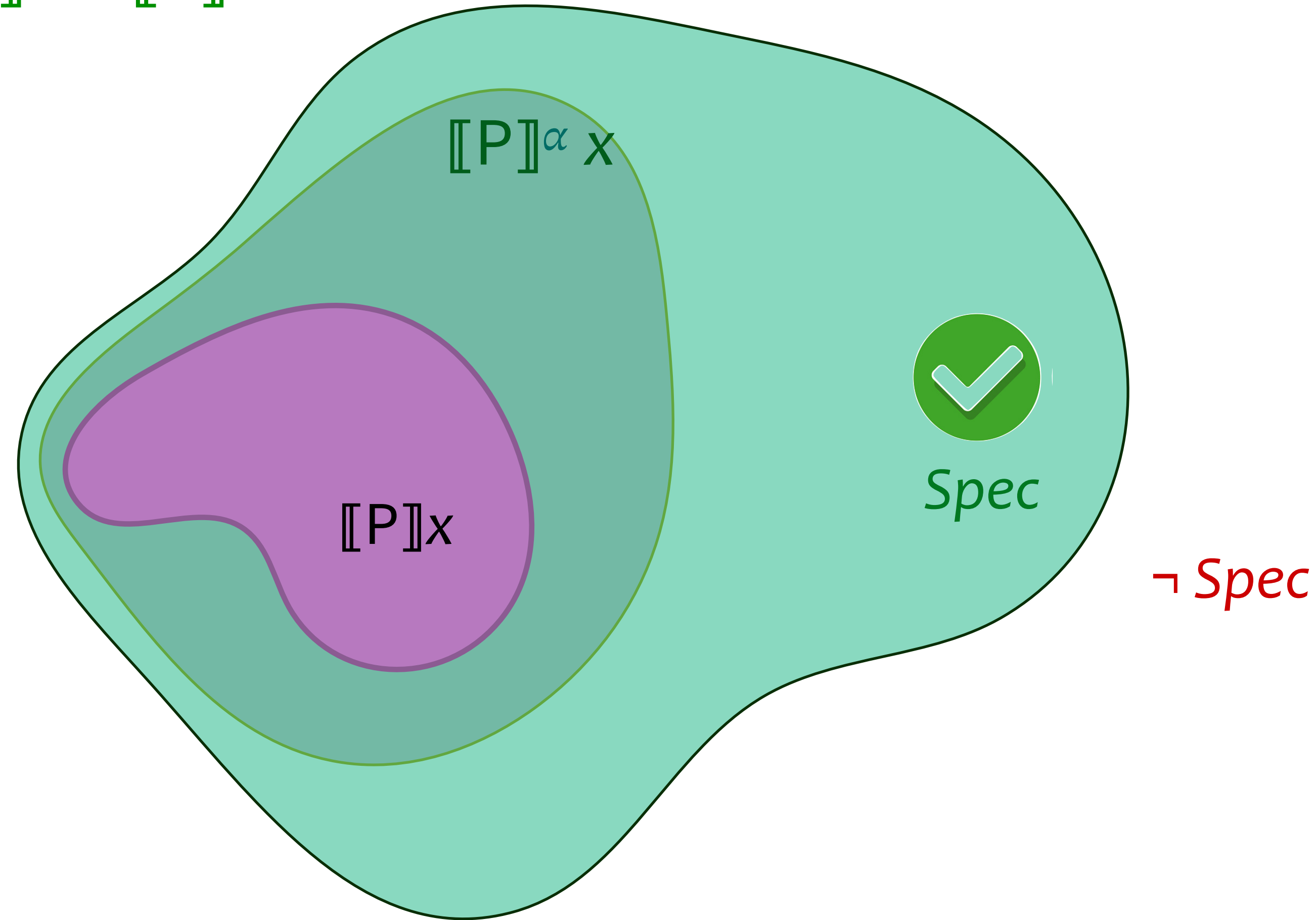
VERIFICATION PROBLEM

► $\llbracket P \rrbracket x \subseteq \text{Spec?}$

$\llbracket P \rrbracket^\alpha x \subseteq \text{Spec?}$

Decidable!

$$\llbracket P \rrbracket x \subseteq \llbracket P \rrbracket^\alpha x$$



Se $\llbracket P \rrbracket x \subseteq \llbracket P \rrbracket^\alpha x \subseteq \text{Spec}$

VERIFICATION PROBLEM

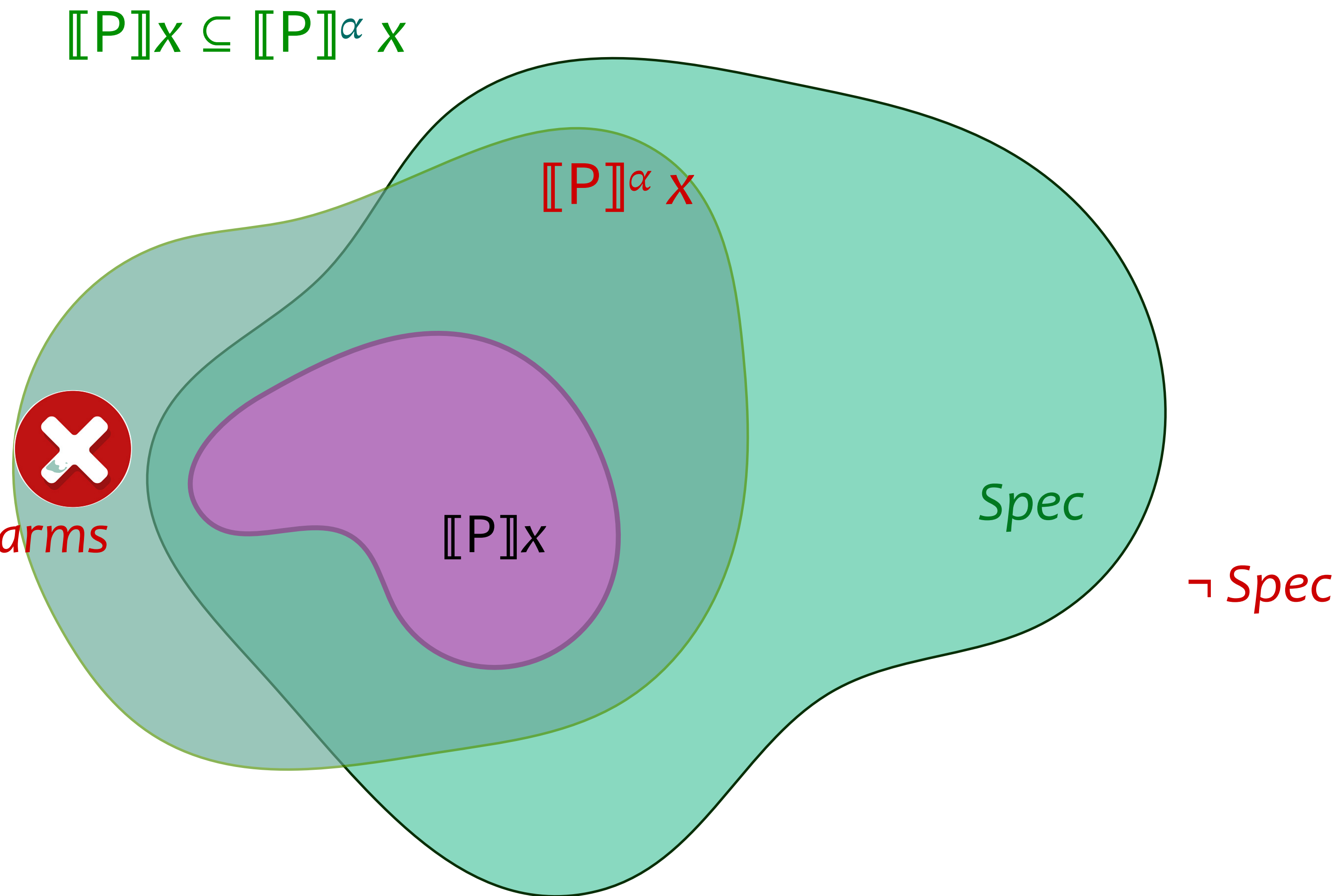
[CC77,CC79]

► $\llbracket P \rrbracket x \subseteq \text{Spec?}$

$\llbracket P \rrbracket^\alpha x \subseteq \text{Spec?}$

Decidable!

False alarms



Se $\llbracket P \rrbracket x \subseteq \alpha(\llbracket P \rrbracket x) \subseteq \llbracket P \rrbracket^\alpha x \not\subseteq \text{Spec}$

ABSTRACTION PROCESS

- We build $\llbracket P \rrbracket^\#$ from $\llbracket P \rrbracket$ inductively on the language syntax
- The semantics $\llbracket P \rrbracket$ is a pair:
 - an (*ordered*) domain of computation D
 - a function f (*with fix point*)
- In the approximation process $\llbracket P \rrbracket^\#$ is also a pair:
 - **Abstraction of data:** Definition of the abstraction $D^\#$ of the domain D of computation and the relations between concrete and abstract objects
 - **Abstraction of computations:** Abstraction $f^\#$ of the computation f with particular attention to the relation with the concrete function f

ABSTRACTING DATA (RECALL)

- *Gl and Moore families*: $C \xrightleftharpoons[\alpha]{\gamma} A$ iff A is isomorphic to a Moore family of C
- *uco and Gl*: Let ρ be an uco on C then there exists an isomorphism $\iota : \rho(C) \rightarrow A$ and
$$C \xrightleftharpoons[\iota \circ \rho]{\iota^{-1}} A$$
- *Gl and uco*: Let $(C, \leq_C) \xrightleftharpoons[\alpha]{\gamma} (A, \leq_A)$ then $\gamma\alpha$ is an uco
- *uco and Moore families*: Let ρ be an uco, the set of its fixpoints $\rho(C)$ is a Moore family of C

UPPER CLOSURE OPERATORS

➤ A function $\rho : P \rightarrow P$ on a poset $\langle P, \leq_P \rangle$ is called *upper closure operator* if

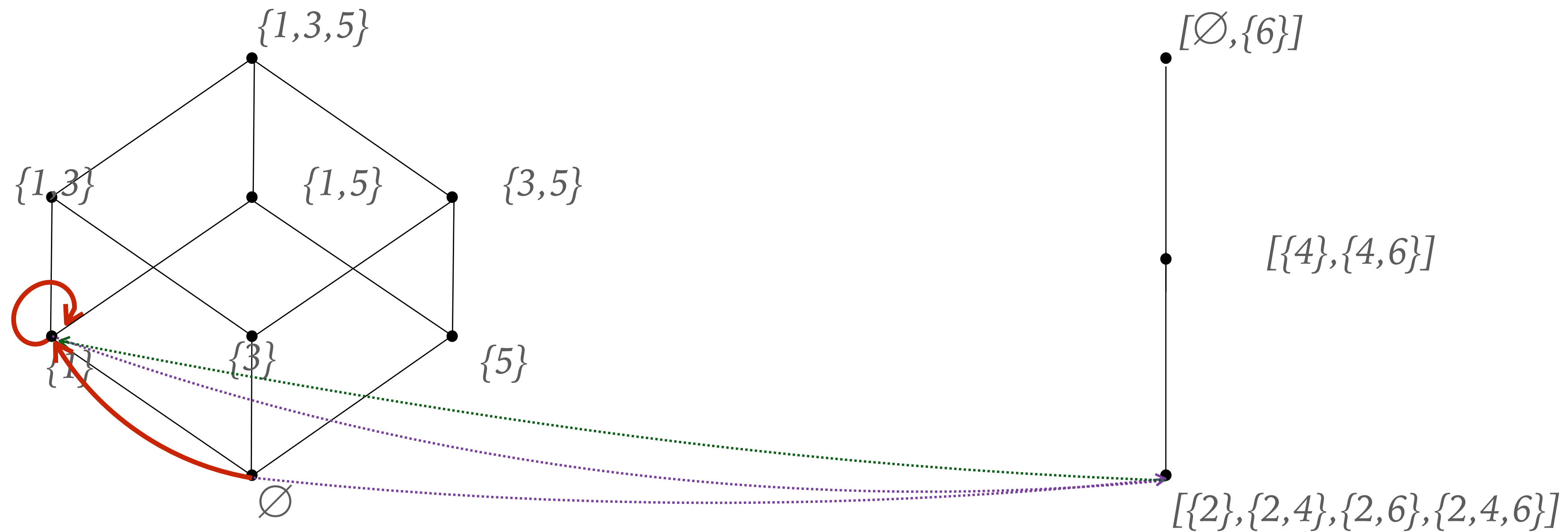
$$(i) \quad \forall x \in P . x \leq_P \rho(x) \quad \textit{Extensive}$$

$$(ii) \quad \forall x, y \in P . (x \leq_P y \Rightarrow \rho(x) \leq_P \rho(y)) \quad \textit{Monotone}$$

$$(iii) \quad \forall x \in P . \rho\rho(x) = \rho(x) \quad \textit{Idempotent}$$

➤ In GC and GI $\gamma\alpha$ is an upper closure operator!

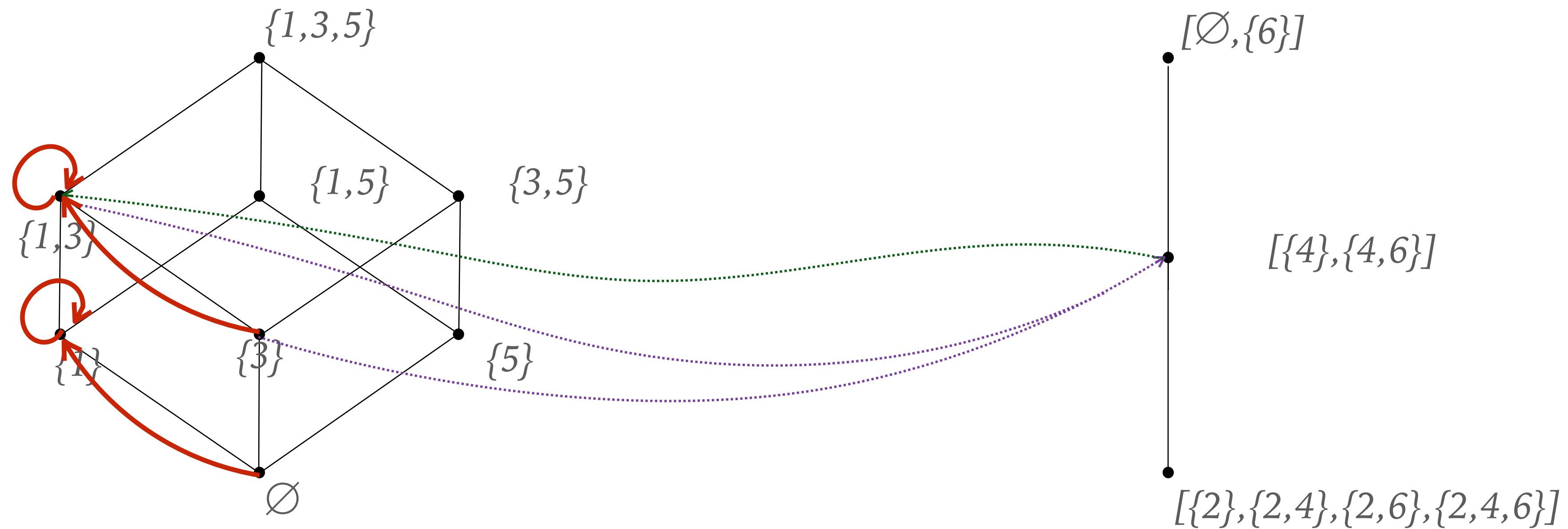
AN EXAMPLE: GI VS UCO



$$\gamma\alpha(A') = \gamma(\{b \in B \mid A' R b\}) = \{a \in A \mid a R \{b \in B \mid A' R b\}\} =$$

$$\{a \in A \mid \forall b \in B. A' R b \text{ then } a R b\}$$

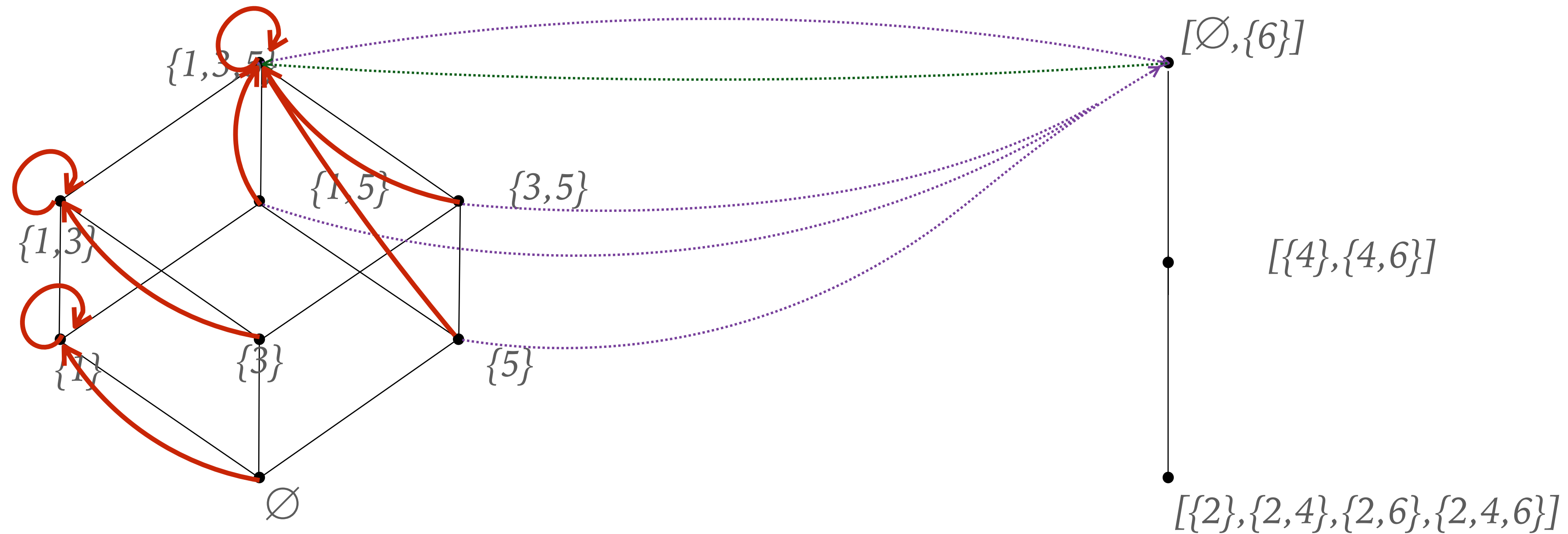
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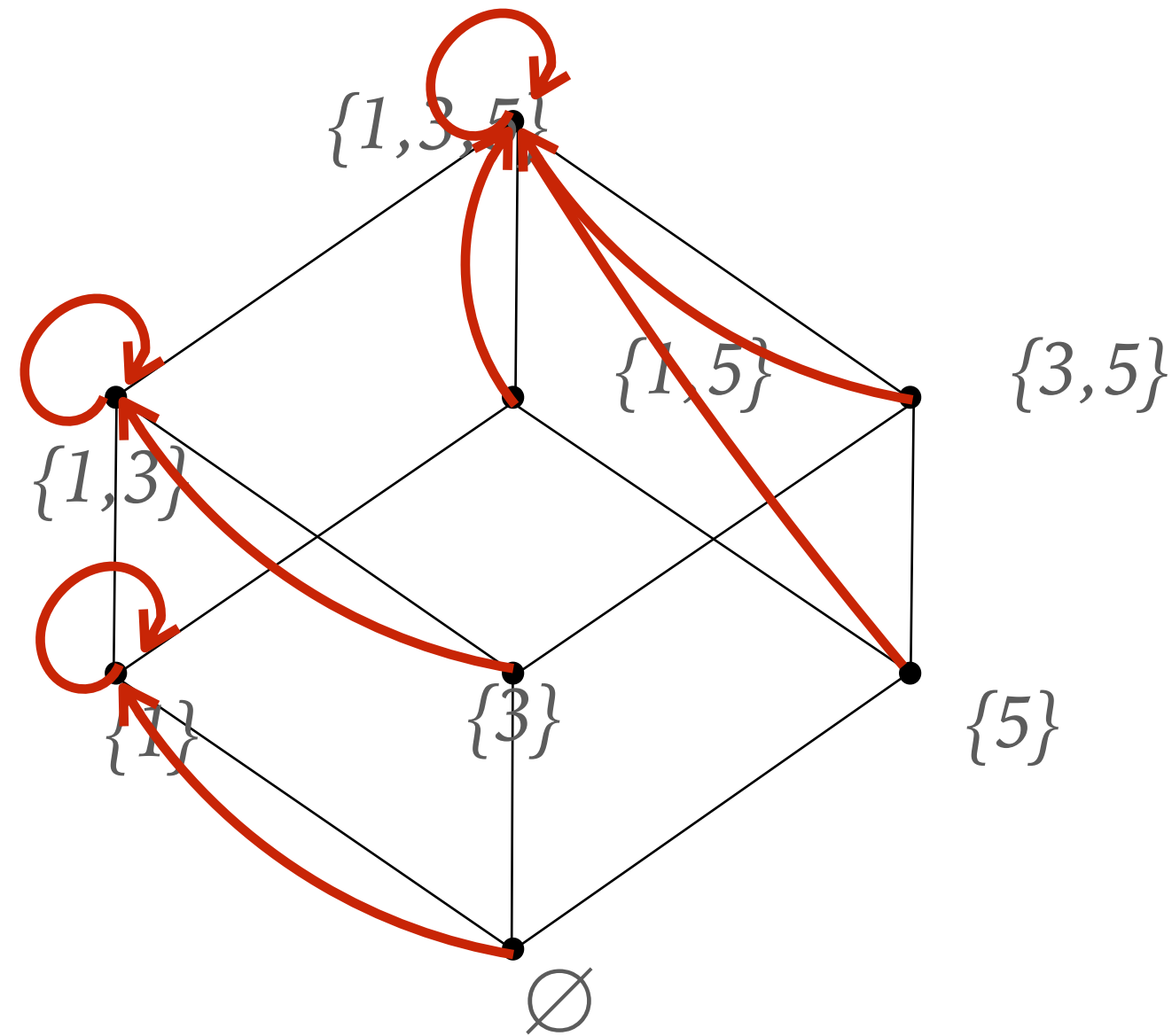
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AN EXAMPLE: UCO



Property characterized Π : A satisfies the Π if all its elements are
< of the same set of elements in $\mathcal{P}(\{2, 4, 6\})$

ABSTRACTING COMPUTATIONS

- Once we have the abstract domain of objects we want to move computation on this domain for gaining decidability but keeping the over-approximation
- Let $C \xrightleftharpoons[\alpha]{\gamma} A$ be a GI, $f : C \rightarrow C$ a concrete operations and let $f^\# : A \rightarrow A$ a possible abstraction. It is said to be *sound* if

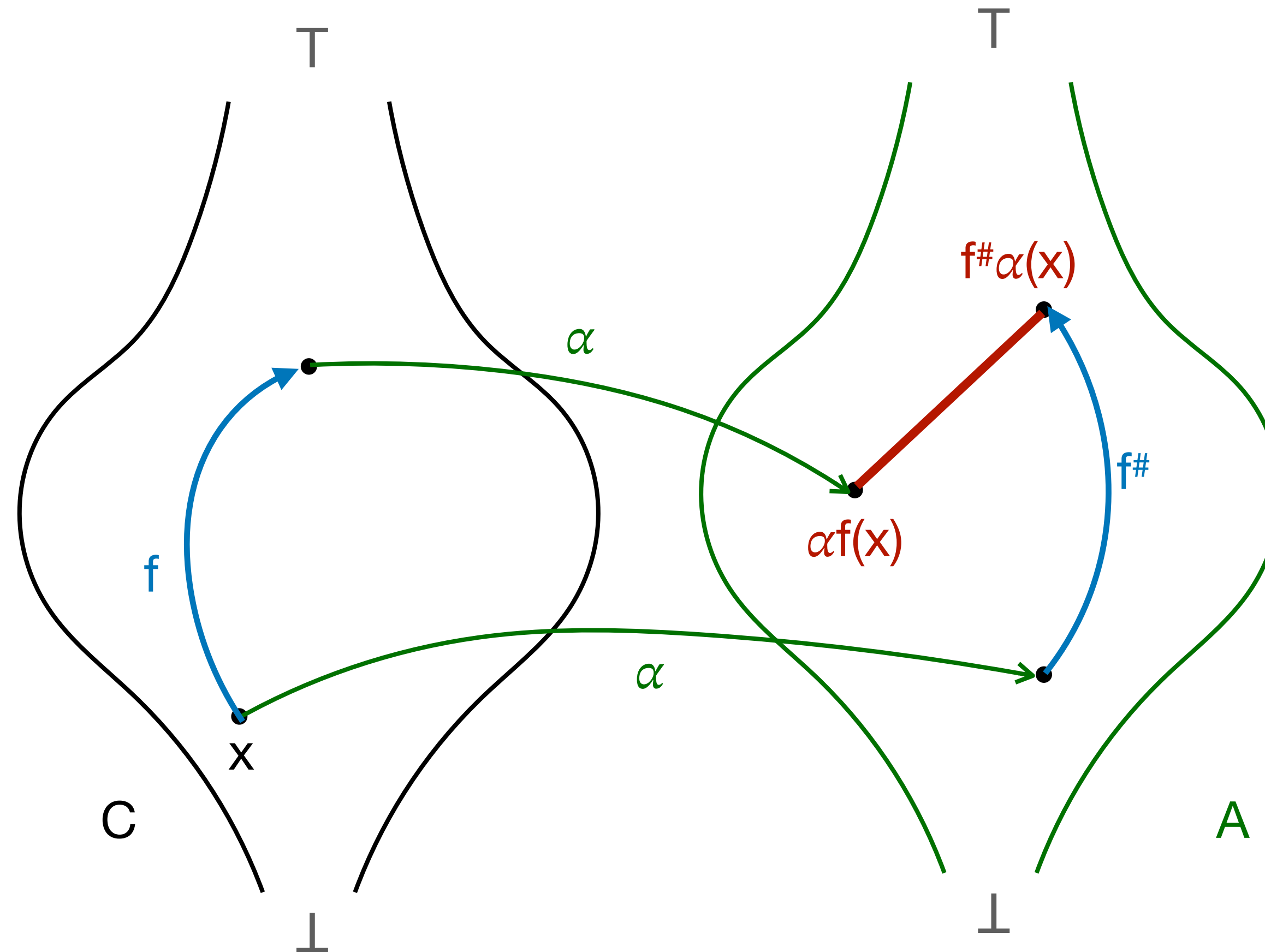
$$\forall x \in C . \alpha \circ f(x) \leq_A f^\# \circ \alpha(x)$$

or equivalently

$$\forall x \in A . f \circ \gamma(x) \leq_C \gamma \circ f^\#(x)$$

CONCRETE VS ABSTRACT COMPUTATION

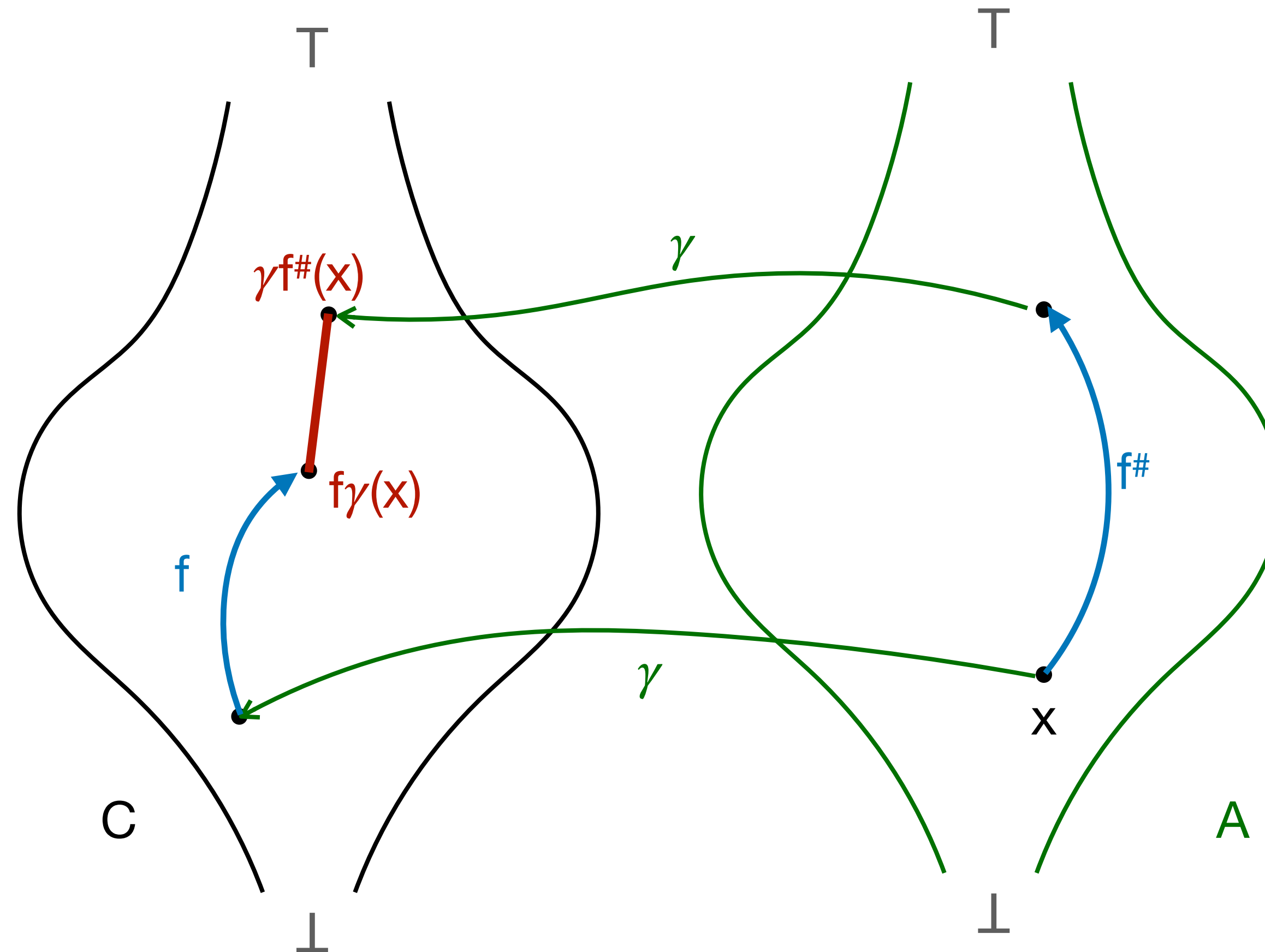
BACKWARD



$$\forall x \in C . \alpha \circ f(x) \leq_A f^\# \circ \alpha(x)$$

CONCRETE VS ABSTRACT COMPUTATION

FORWARD

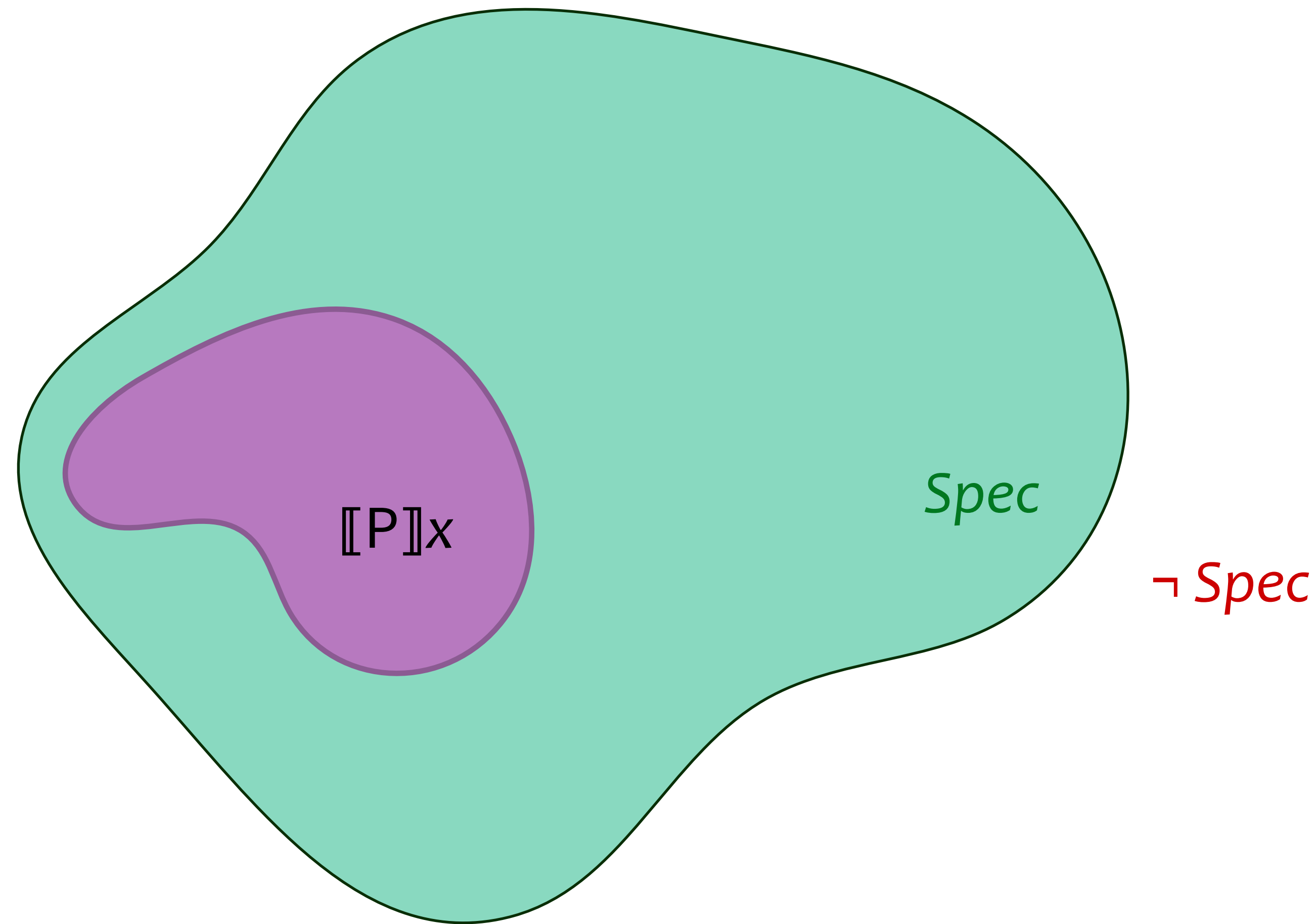


$$\forall x \in A . f \circ \gamma(x) \leq_C \gamma \circ f^\#(x)$$

SOUNDNESS IN PROGRAM SEMANTICS

[CC77,CC79]

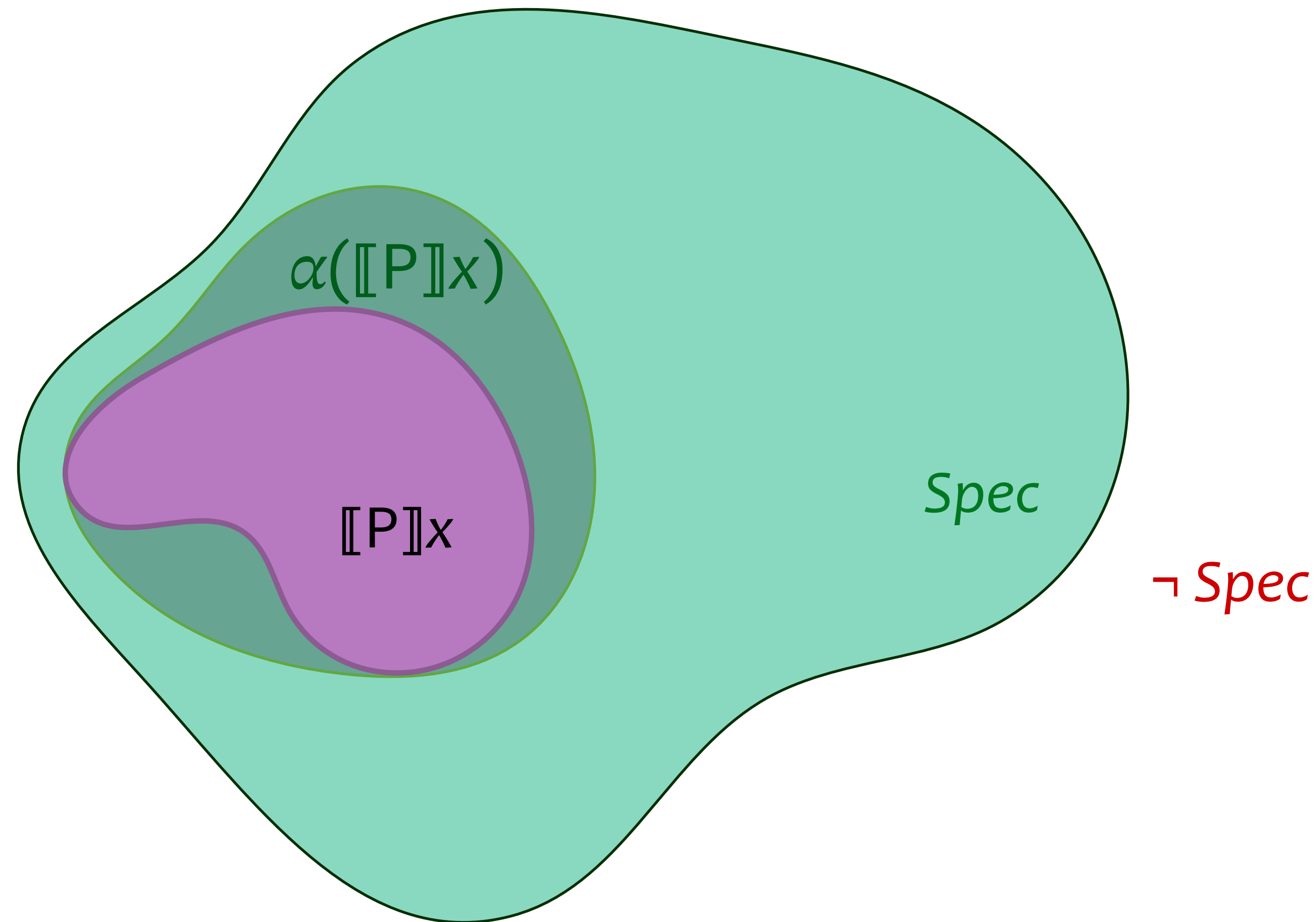
► $\llbracket P \rrbracket x \subseteq \text{Spec?}$



SOUNDNESS IN PROGRAM SEMANTICS

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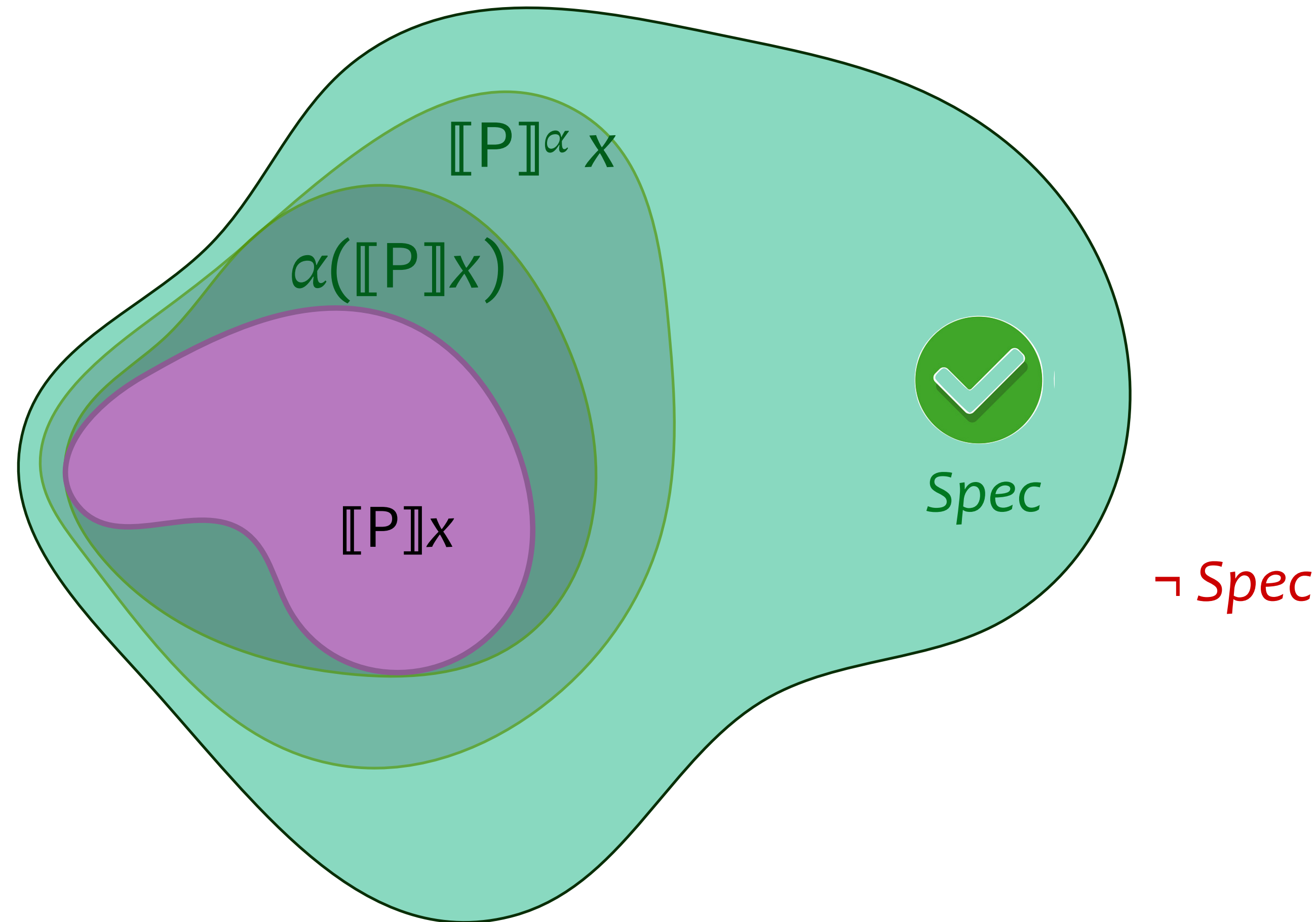


$$\llbracket P \rrbracket x \subseteq \alpha(\llbracket P \rrbracket x)$$

SOUNDNESS IN PROGRAM SEMANTICS

[CC77,CC79]

► $\llbracket P \rrbracket x \subseteq \text{Spec}$?



$$\llbracket P \rrbracket x \subseteq \alpha(\llbracket P \rrbracket x) \subseteq \llbracket P \rrbracket^\alpha x \subseteq \text{Spec}$$

BEST CORRECT APPROXIMATION

- Let $C \xleftrightarrow[\alpha]{\gamma} A$ be a GI, $f : C \rightarrow C$ be a concrete function and $f^\# : A \rightarrow A$ an abstract function. We have that

$$\forall x \in C. \alpha \circ f(x) \leq_A f^\# \circ \alpha(x) \Leftrightarrow \forall x \in A. \alpha \circ f \circ \gamma(x) \leq_A f^\#(x)$$

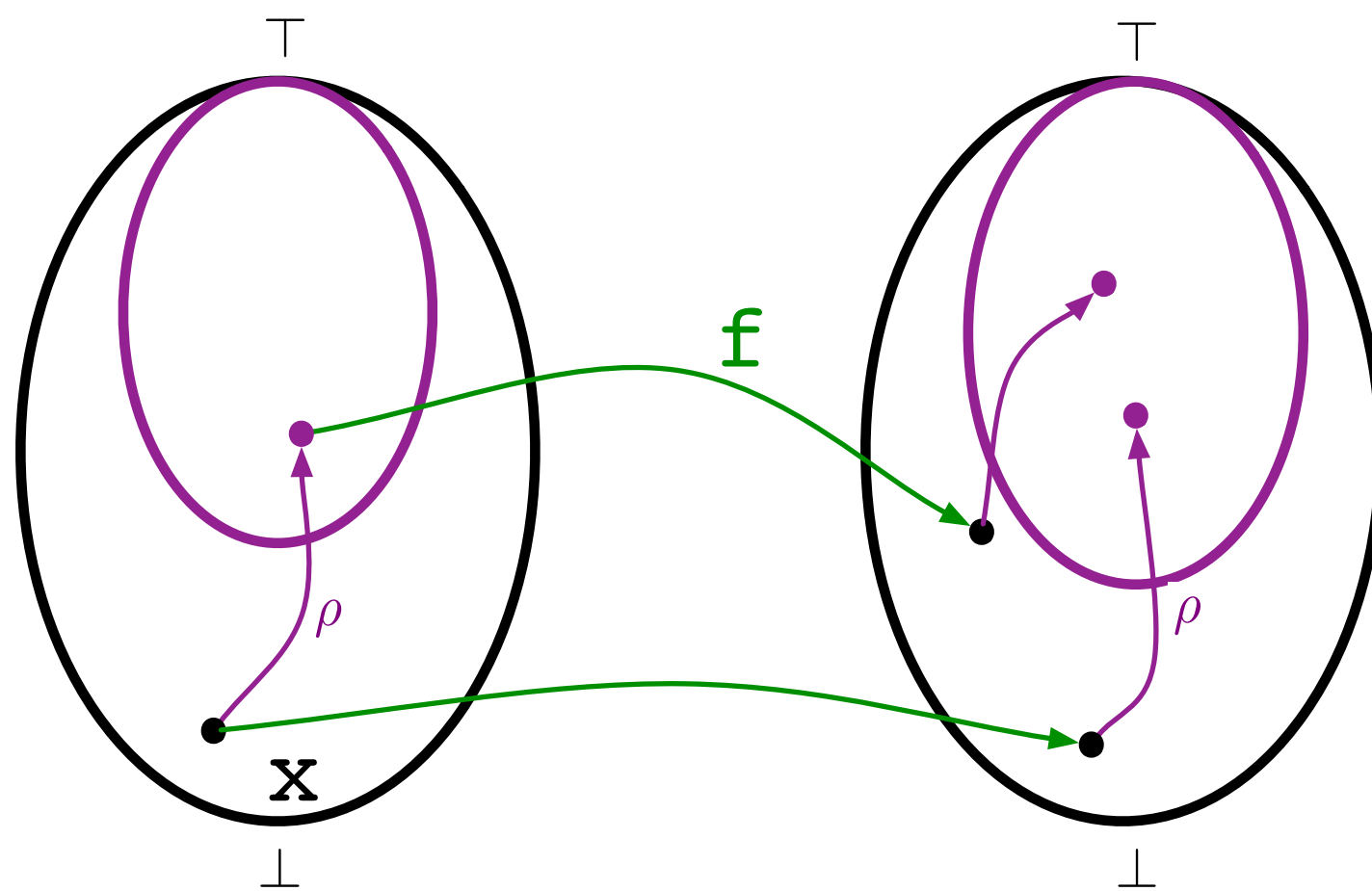
- As a consequence we have that $\alpha \circ f \circ \gamma : A \rightarrow A$ is the *best correct approximation (bca)* of $f : C \rightarrow C$ in the abstract domain
- For the whole semantics is useless, since it passes through the concrete computation

SOUNDNESS AS DOMAIN PROPERTY

- If $\alpha \circ f \circ \gamma(x) = f^\#(x)$
- Then backward soundness $\forall x \in C . \alpha \circ f(x) \leq_A f^\# \circ \alpha(x)$ becomes $\rho \circ f \circ \rho \geq \rho \circ f$
- While forward soundness $\forall x \in A . f \circ \gamma(x) \leq_C \gamma \circ f^\#(x)$ becomes $\rho \circ f \circ \rho \geq f \circ \rho$
- THESE FORMS OF SOUNDNESS ARE EQUIVALENT!
 - Backward soundness holds iff forward soundness holds.

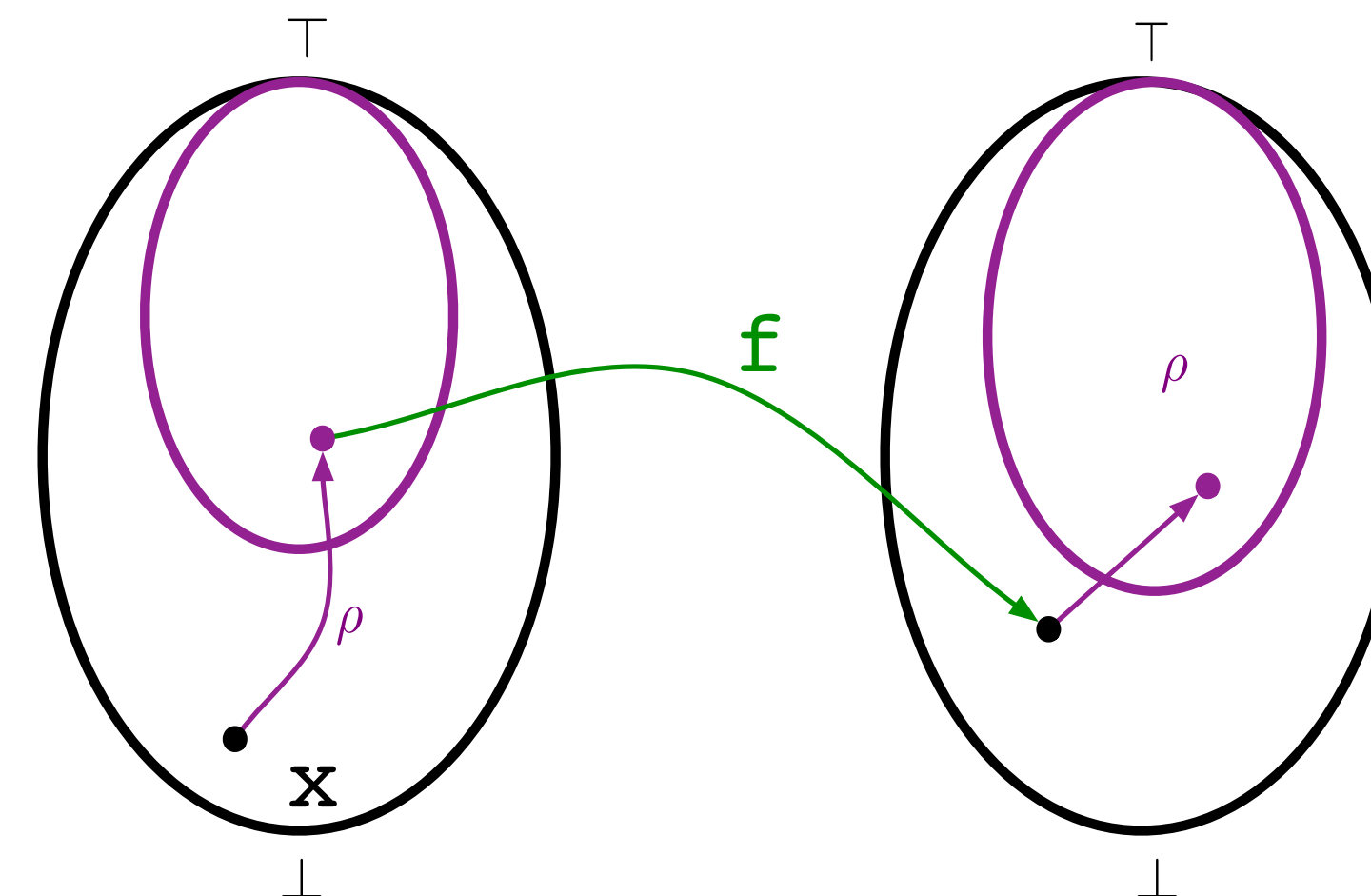
...ON CLOSURES

BACKWARD



$$\rho \circ f \circ \rho \geq \rho \circ f$$

FORWARD

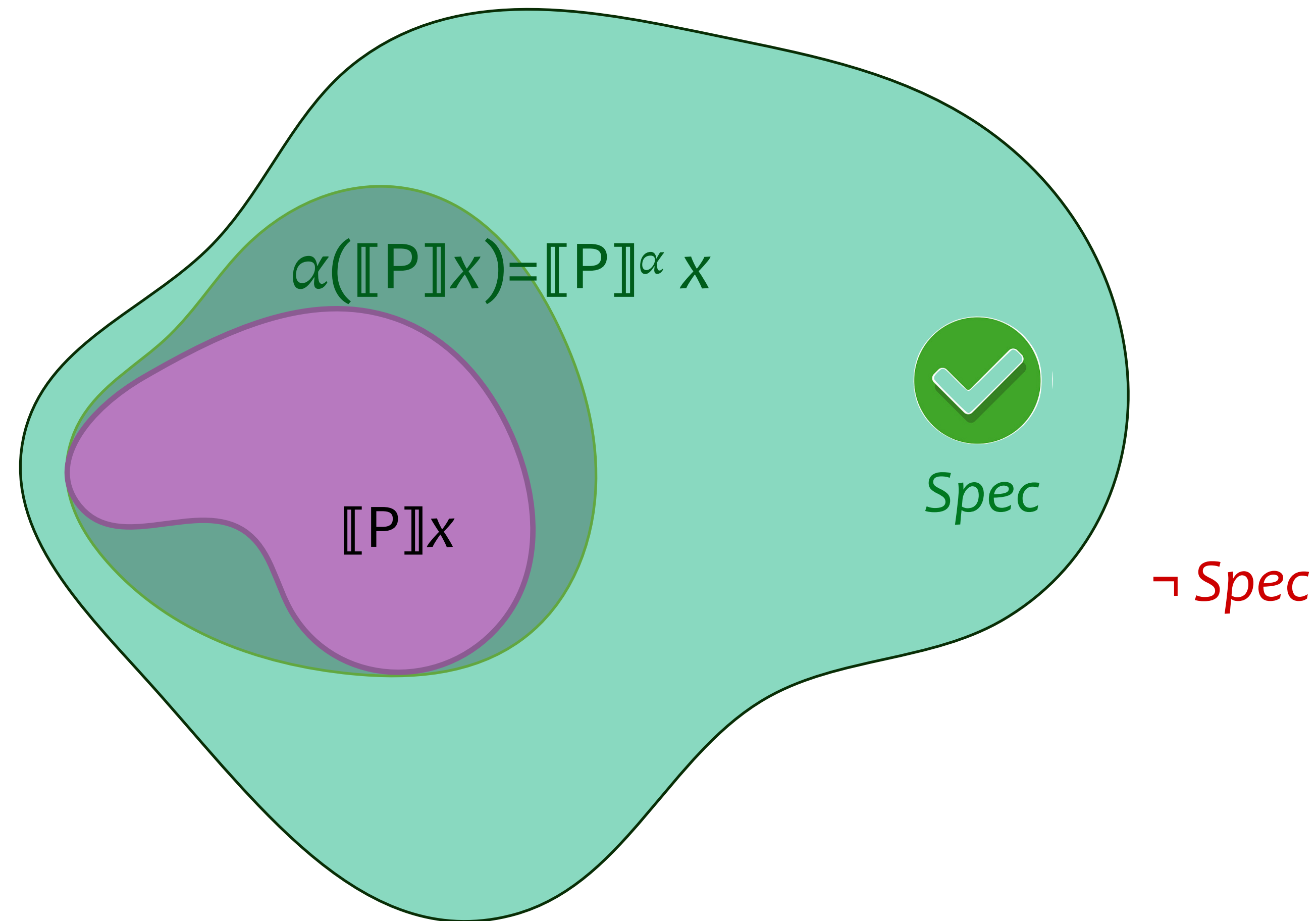


$$\rho \circ f \circ \rho \geq f \circ \rho$$

ABSTRACT INTERPRETATION MAY BE COMPLETE

[CC77,CC79]

- $\llbracket P \rrbracket x \subseteq \text{Spec}$?
- $\alpha(\text{Spec}) = \text{Spec}$



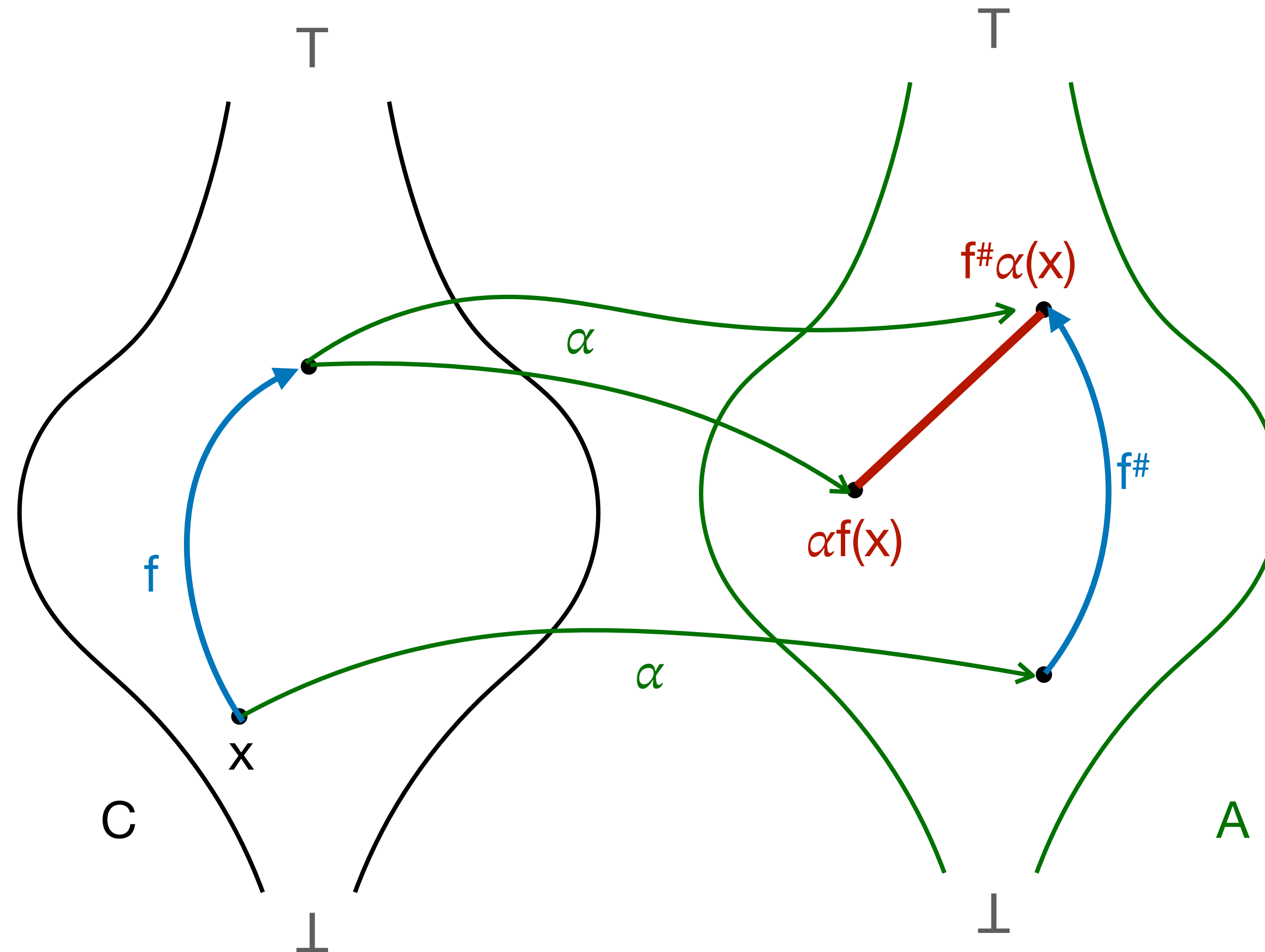
$$\llbracket P \rrbracket x \subseteq \alpha(\llbracket P \rrbracket x) = \llbracket P \rrbracket^\alpha x \subseteq \text{Spec}$$

COMPLETENESS

- Completeness encode an ideal situation where no loss of precision arises in abstract computation
- When we strength the soundness conditions by requiring *equality* we obtain two incomparable notions of completeness characterizing precision.
 - Backward completeness considers abstractions on the output of operations
 - Forward completeness considers abstractions on the input to operations

CONCRETE VS ABSTRACT COMPUTATION

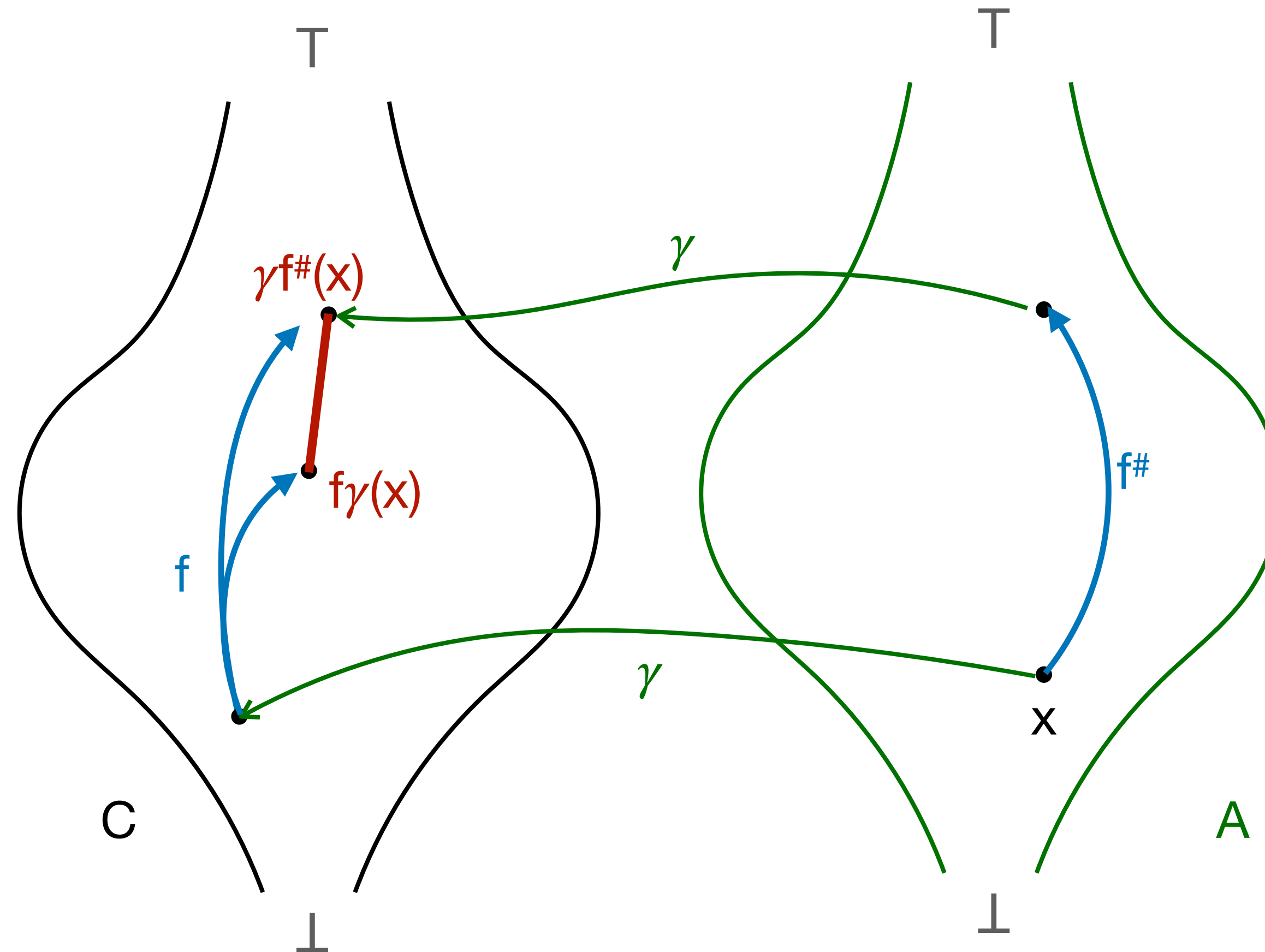
BACKWARD



$$\forall x \in C . \alpha \circ f(x) = f^{\#} \circ \alpha(x)$$

CONCRETE VS ABSTRACT COMPUTATION

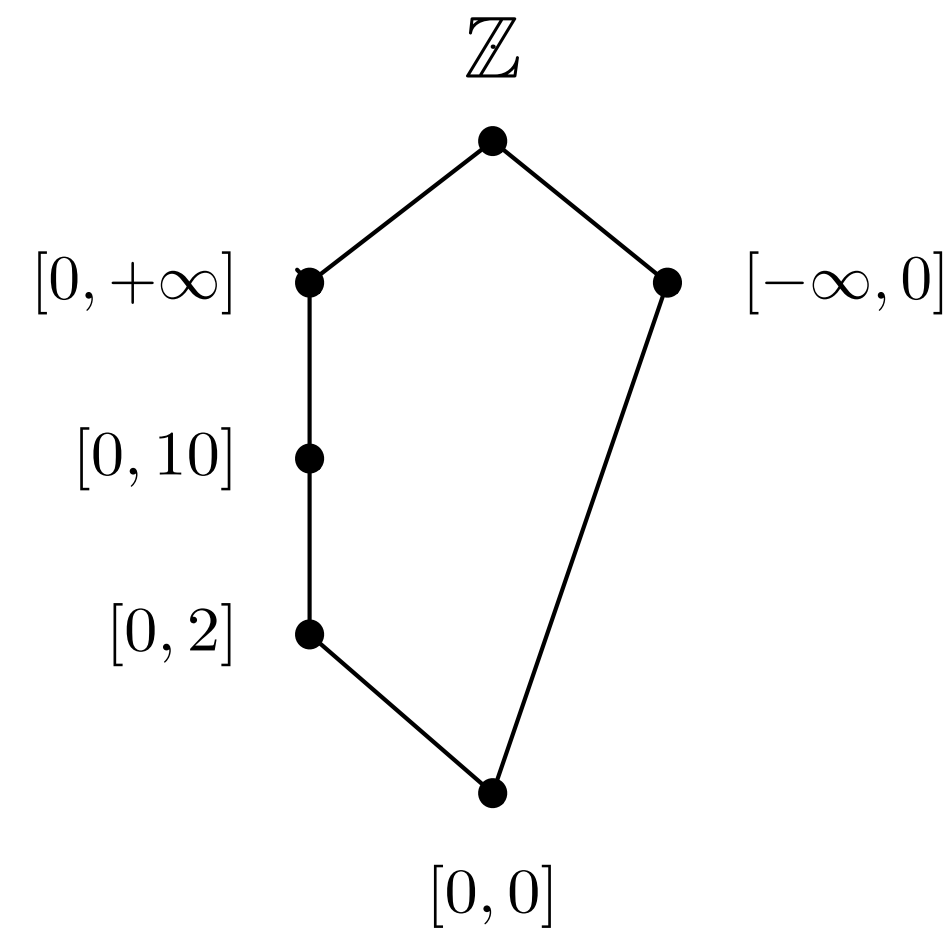
FORWARD



$$\forall x \in A . f \circ \gamma(x) = \gamma \circ f^\#(x)$$

A CLASSICAL SIMPLE EXAMPLE

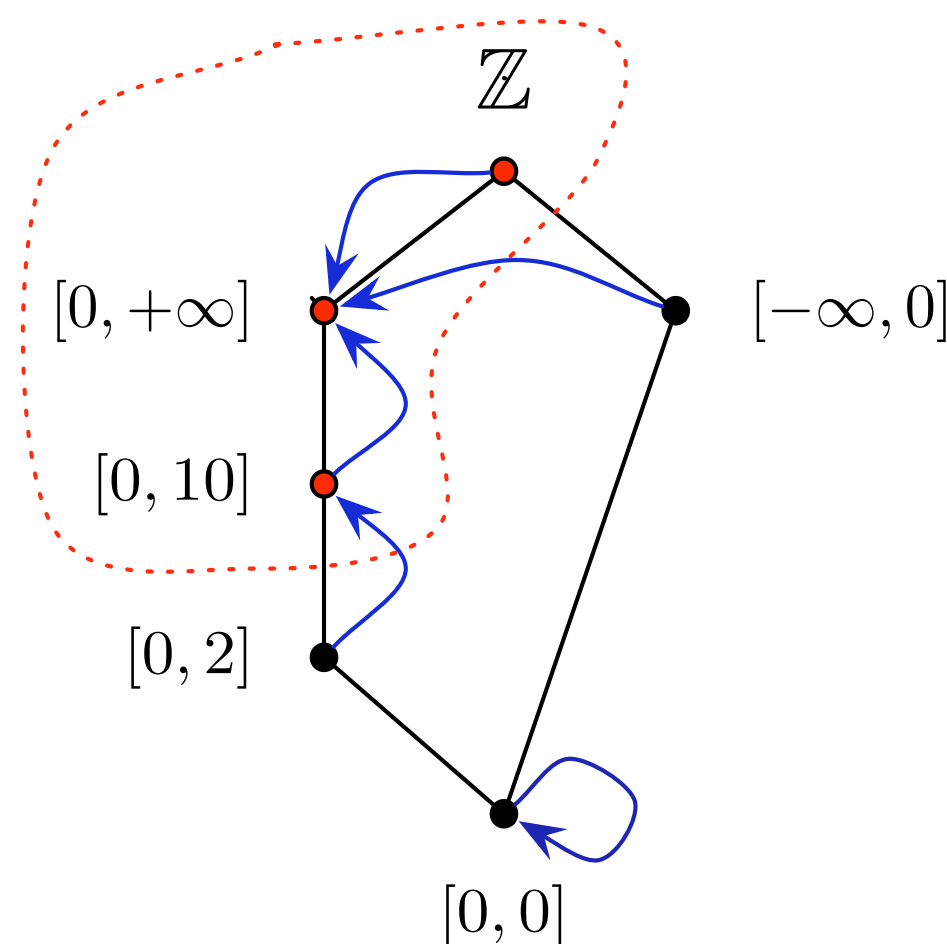
- Consider the domain of intervals as concrete (depicted a portion)
- Consider the concrete function $\text{sq}([a,b])=[a^2,b^2]$



A CLASSICAL SIMPLE EXAMPLE

- Consider the domain of intervals as concrete (depicted a portion)
- Consider the concrete function $\text{sq}([a,b])=[a^2,b^2]$

A is not backward complete in $[0,2]$



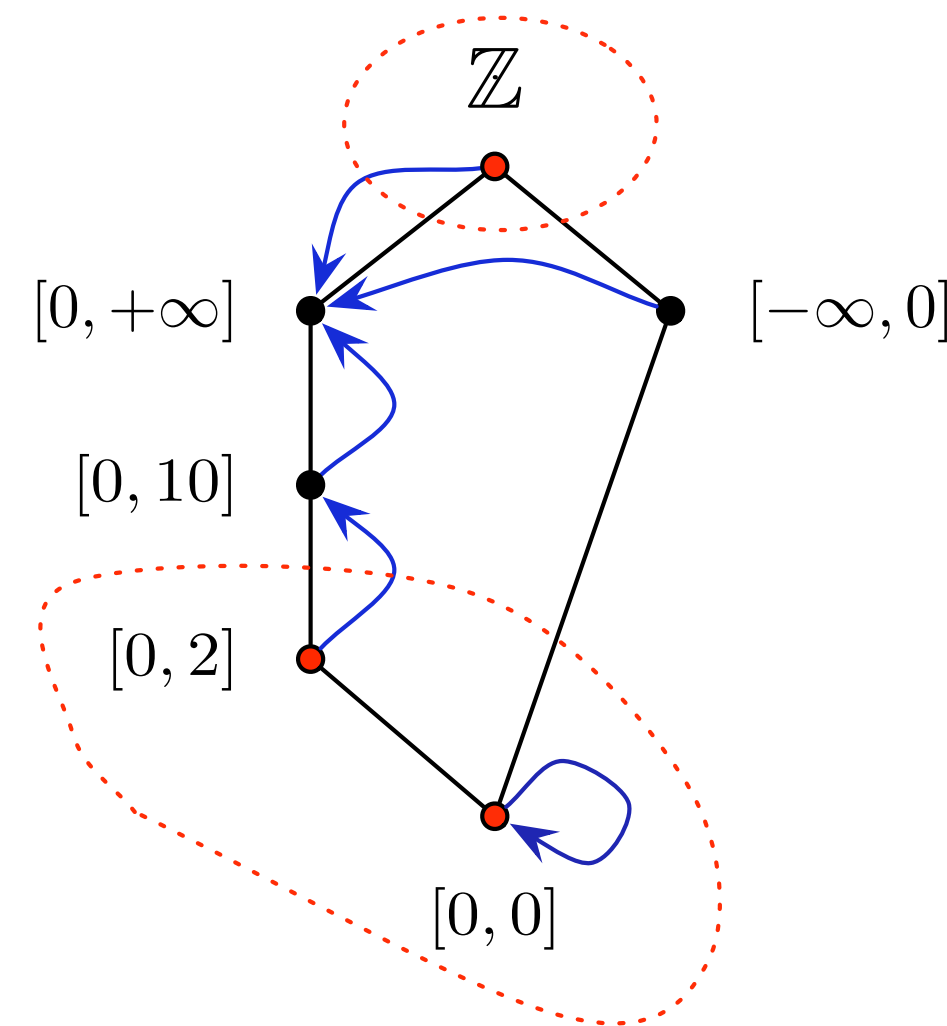
$$\alpha \circ \text{sq}([0,2]) = \alpha([0,4]) = [0,10]$$
$$\text{sq}^\# \circ \alpha([0,2]) = \text{sq}^\#([0,10]) = [0,100]$$

- Consider the abstraction $A=\{\mathbb{Z},[0,+\infty],[0,10]\}$ (abstraction function denoted α)
- Let the abstract function $\text{sq}^\#([a,b]) = \alpha \circ \text{sq}([a,b])$

A CLASSICAL SIMPLE EXAMPLE

- Consider the domain of intervals as concrete (depicted a portion)
- Consider the concrete function $\text{sq}([a,b])=[a^2,b^2]$

A is not forward complete in $[0,2]$



$$\begin{aligned}\text{sq} \circ \gamma([0,2]) &= \text{sq}([0,2]) = [0,4] \\ \gamma \circ \text{sq}^\#([0,2]) &= \gamma([0,10]) = [0,10]\end{aligned}$$

- Consider the abstraction $A=\{\mathbb{Z},[0,2],[0,0]\}$ (concretization function denoted $\gamma=\text{id}$)
- Let the abstract function $\text{sq}^\#([a,b]) = \alpha \circ \text{sq}([a,b])$

BACKWARD COMPLETENESS

- Backward completeness means that we do not lose precision due to the input abstraction
 - Namely, even if we approximate the input the semantics is still able to compute the precise abstract property
 - The added input noise is not disturbing the semantics computation
- Examples of incompleteness:
 - Sign for sum : Pos + Neg = I don't know
 - The information we lose observing sign of inputs is enough to compromise the addition computation.
Backward incomplete: we can split the input observation observing intervals.
 - There is no way to change the output observation in order to get a more precise answer. *Forward complete!*

FORWARD COMPLETENESS

- Forward completeness means that we do not lose precision due to the output abstraction
 - Namely, even if we approximate the output the abstract observation of the semantic computation is still precise
- Examples of incompleteness:
 - Suppose to have the non deterministic expression $e1 \sqcup e2$, which is $e1$ or $e2$ non deterministically, and constant propagation on integers
 - Const for conditional expr : $\text{const1} \sqcup \text{const2} = \text{I don't know}$
 - What we lose observing constant of outputs compromise the observation. *Forward incomplete: Adding all the possible sets of cardinality two to the constant domain gets completeness*
 - There is no way of changing the input observation for getting more precise answers . *Backward complete!*

FORWARD VS BACKWARD

- Example with both incompleteness:
 - Suppose $\text{Type} = \{\text{Int}, \text{Float}, \text{String}\}$ where strings are interpreted as the longest numerical prefix (integer or Float) when used in arithmetic operations
 - Type for sum : Integer + String = I don't know
 - What we lose observing type of inputs is enough to compromise the addition computation. *Backward incomplete: Splitting strings depending on their numerical prefix gets completeness*
 - BUT, also the output observation adds noise. *Forward incomplete: Adding the point Num=IntuFloat gets completeness*

COMPLETENESS AS A DOMAIN PROPERTY

- Let $C \xrightleftharpoons[\alpha]{\gamma} A$ be a GI, $f : C \rightarrow C$ be a concrete function, there exists a complete abstract function $f^\sharp : A \rightarrow A$ iff the bca $\alpha \circ f \circ \gamma : A \rightarrow A$ is complete for $f : C \rightarrow C$
- This allows us to characterize completeness as a property depending only on the function and the uco

$$\rho \in uco(C)$$

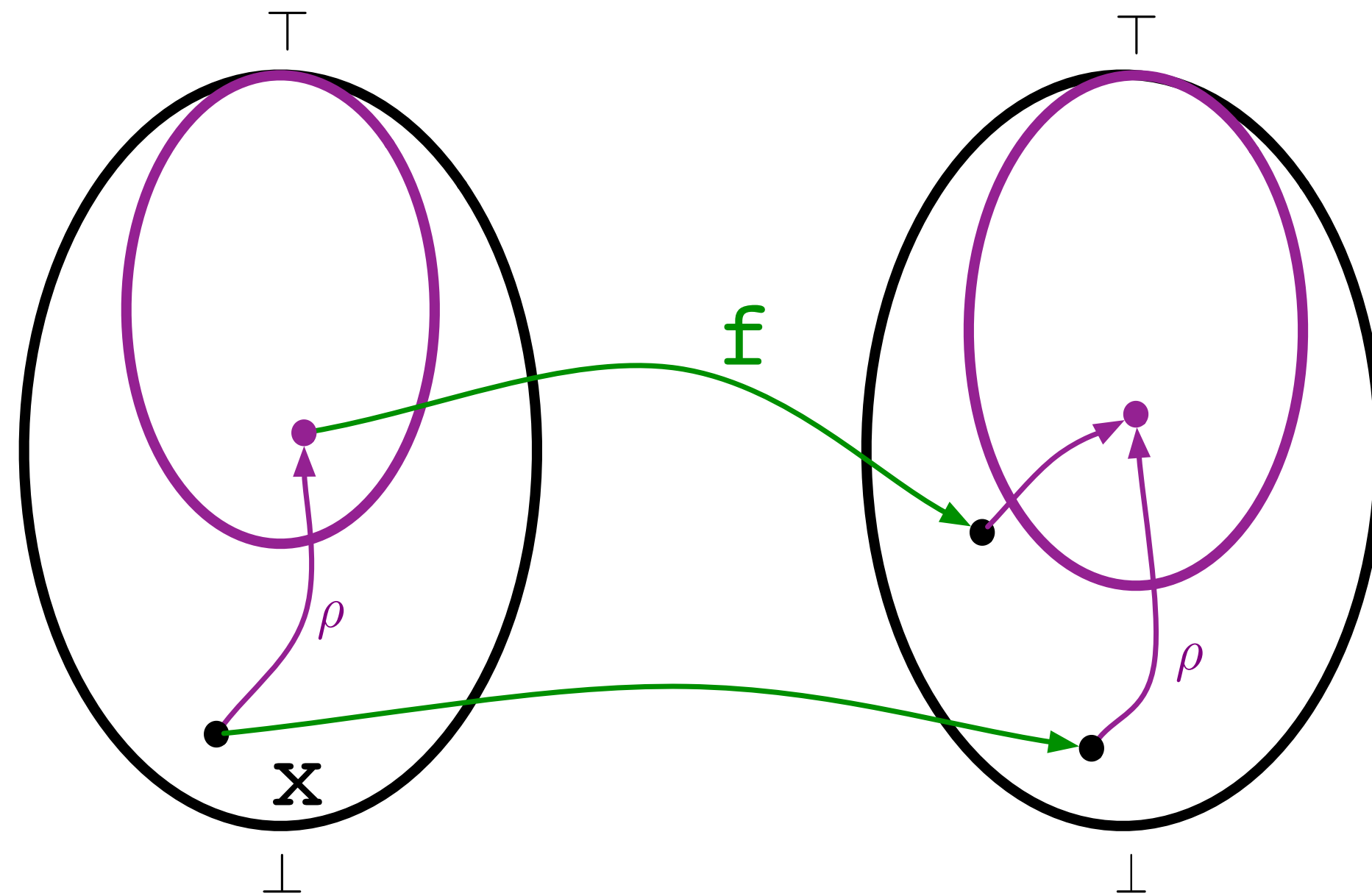
- Backward: $\rho \circ f = \rho \circ f \circ \rho$

- Forward: $f \circ \rho = \rho \circ f \circ \rho$

It is a domain property!

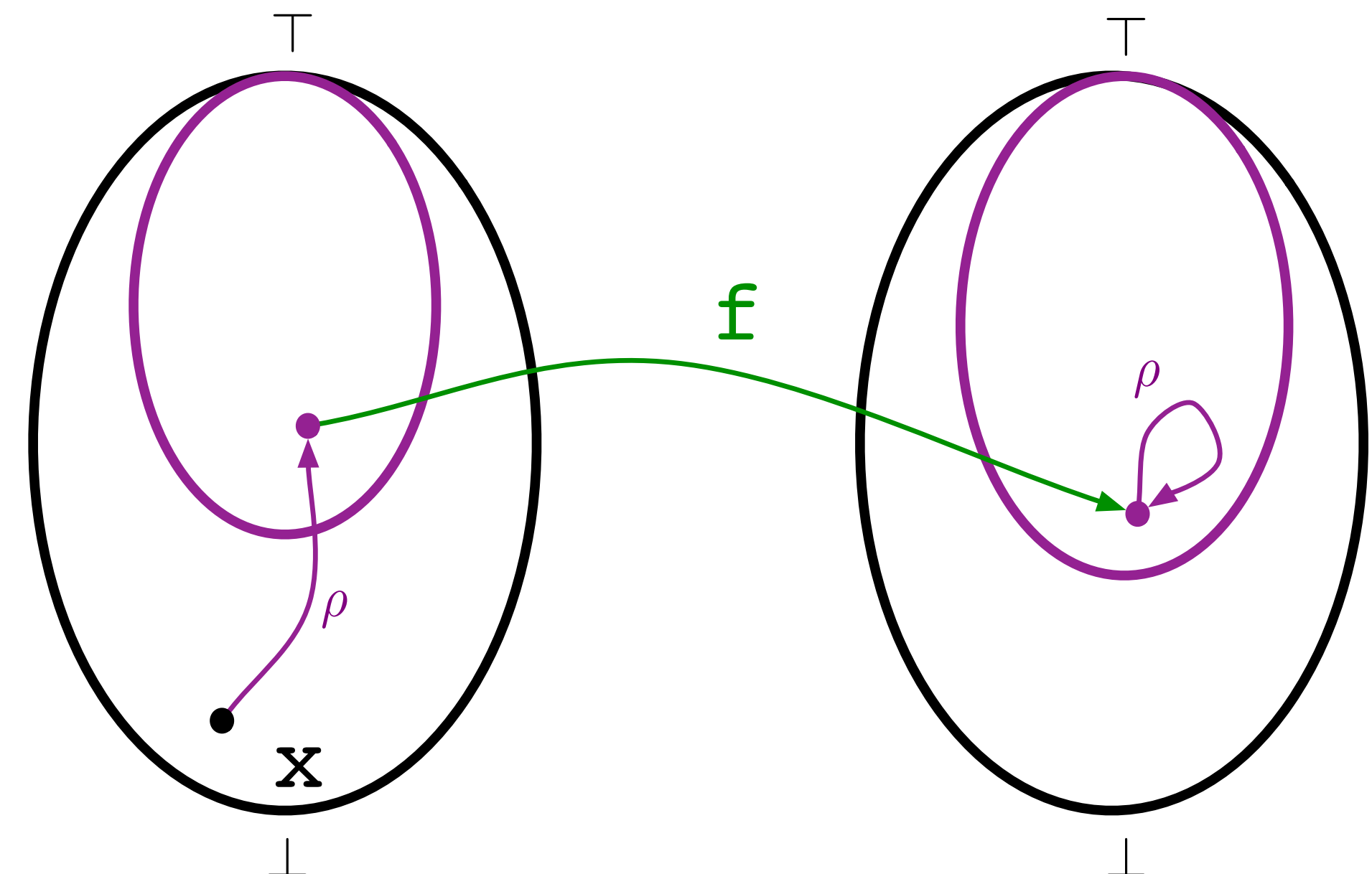
COMPLETENESS AS A DOMAIN PROPERTY

BACKWARD



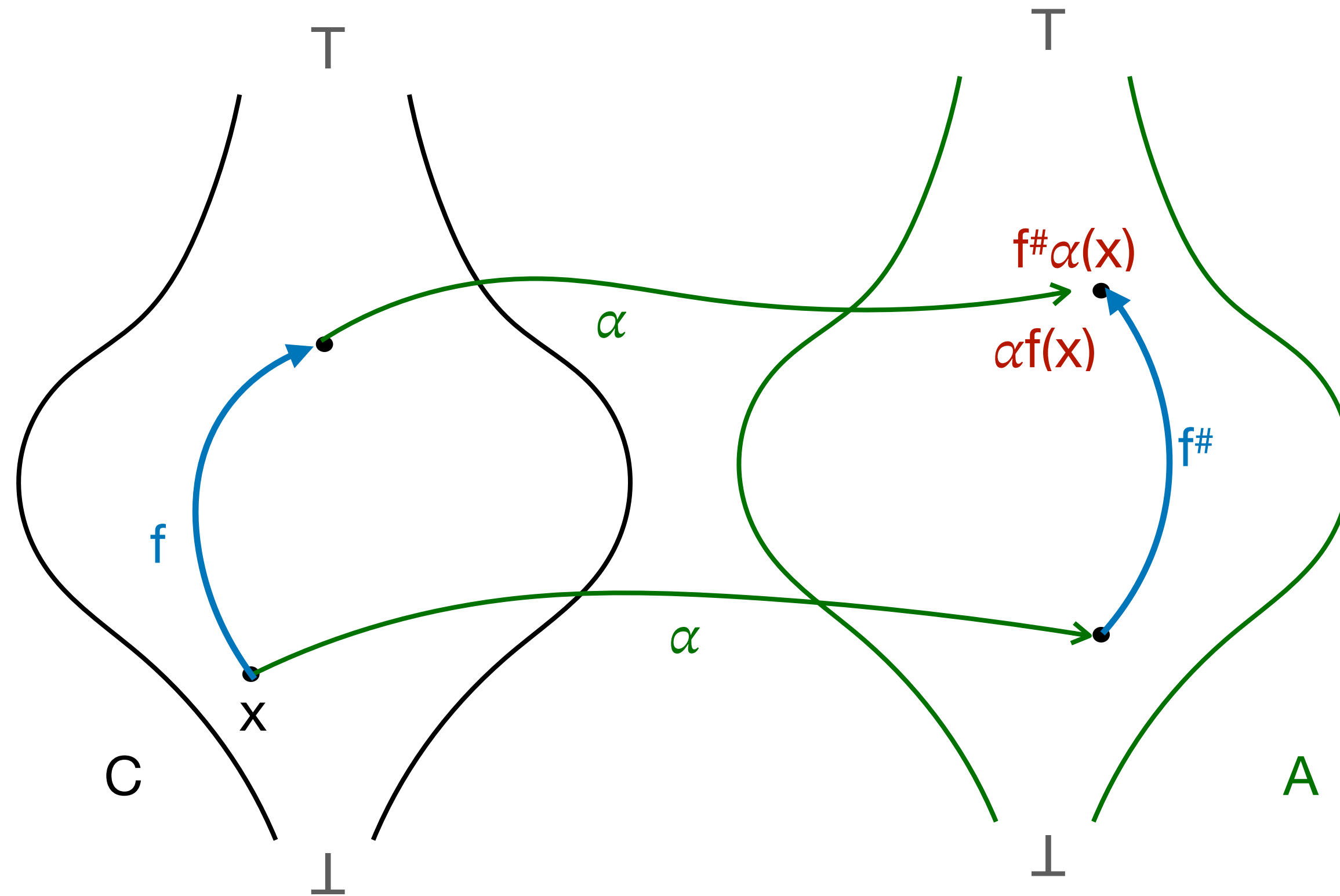
$$\rho \circ f = \rho \circ f \circ \rho$$

FORWARD



$$f \circ \rho = \rho \circ f \circ \rho$$

(BACKWARD) LOCAL COMPLETENESS



Given x (maybe in X)

$$\alpha \circ f(x) = f^{\#} \circ \alpha(x)$$

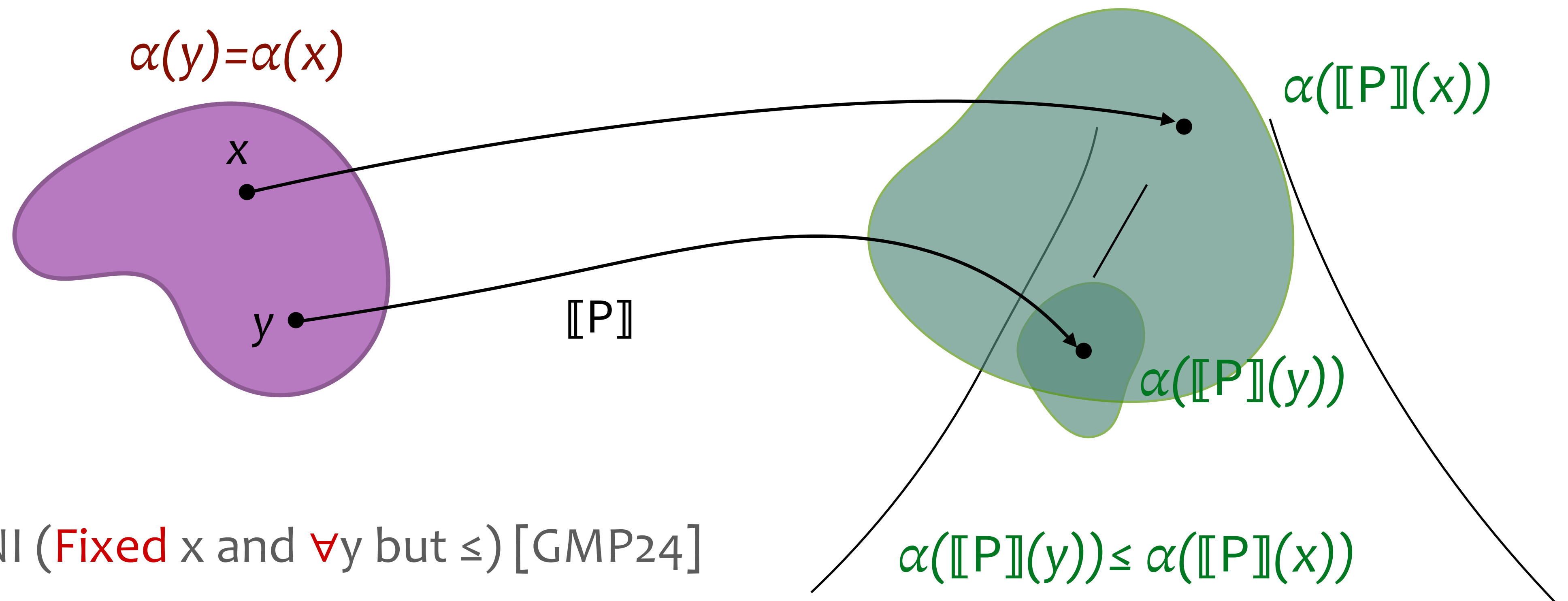
As domain property (on bca)

$$\alpha \circ f(x) = \alpha \circ f \circ \gamma \circ \alpha(x)$$

$$\rho \circ f(x) = \rho \circ f \circ \rho(x)$$

COMPLETENESS AS HYPERPROPERTY

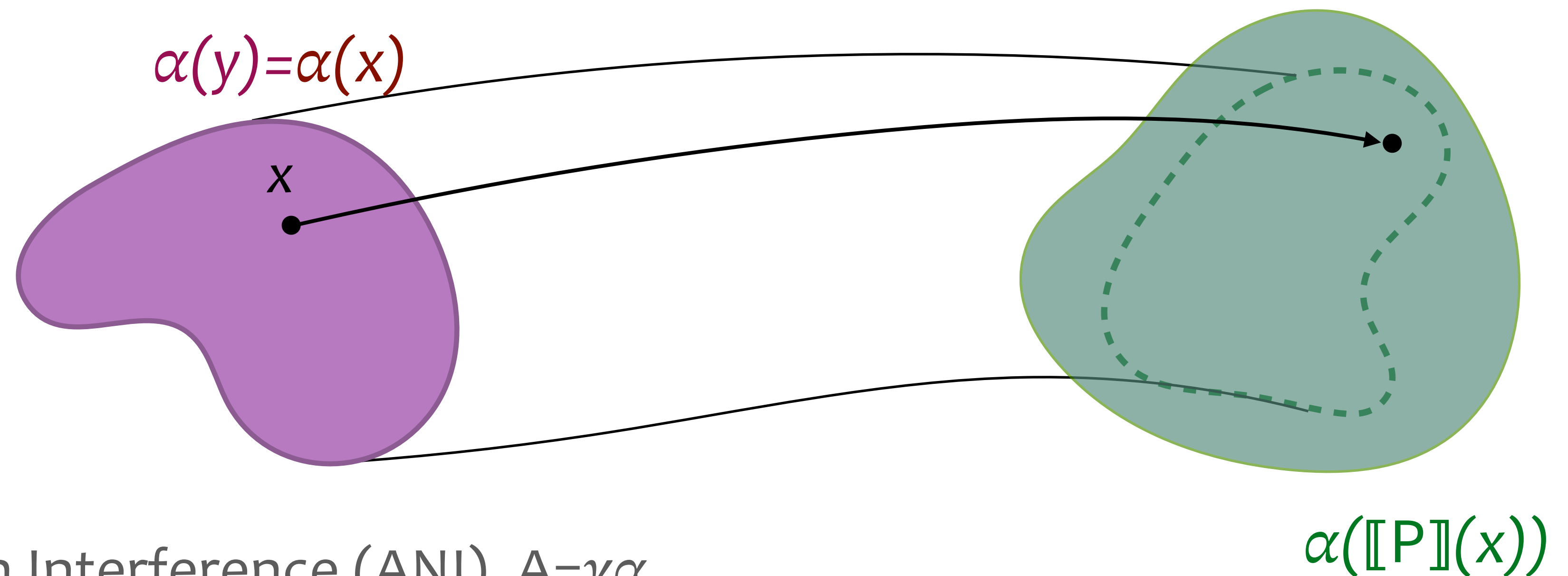
Local Completeness (Fixed x) [BGGR21]



➤ *Relaxed* ANI (Fixed x and $\forall y$ but $\leq$) [GMP24]

COMPLETENESS AS HYPERPROPERTY

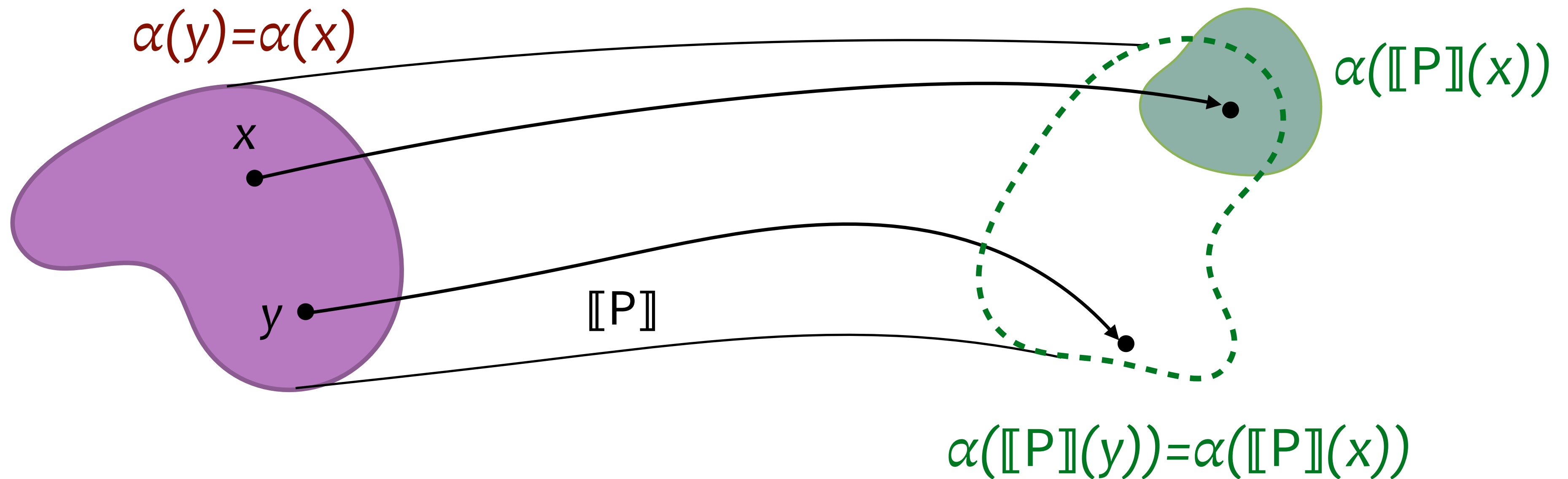
- What if we require the equality?



- **Localizing** Abstract Non Interference (ANI), $A = \gamma\alpha$

COMPLETENESS AS HYPERPROPERTY

➤ Local Abstract Completeness (Fixed $\alpha(x)$)

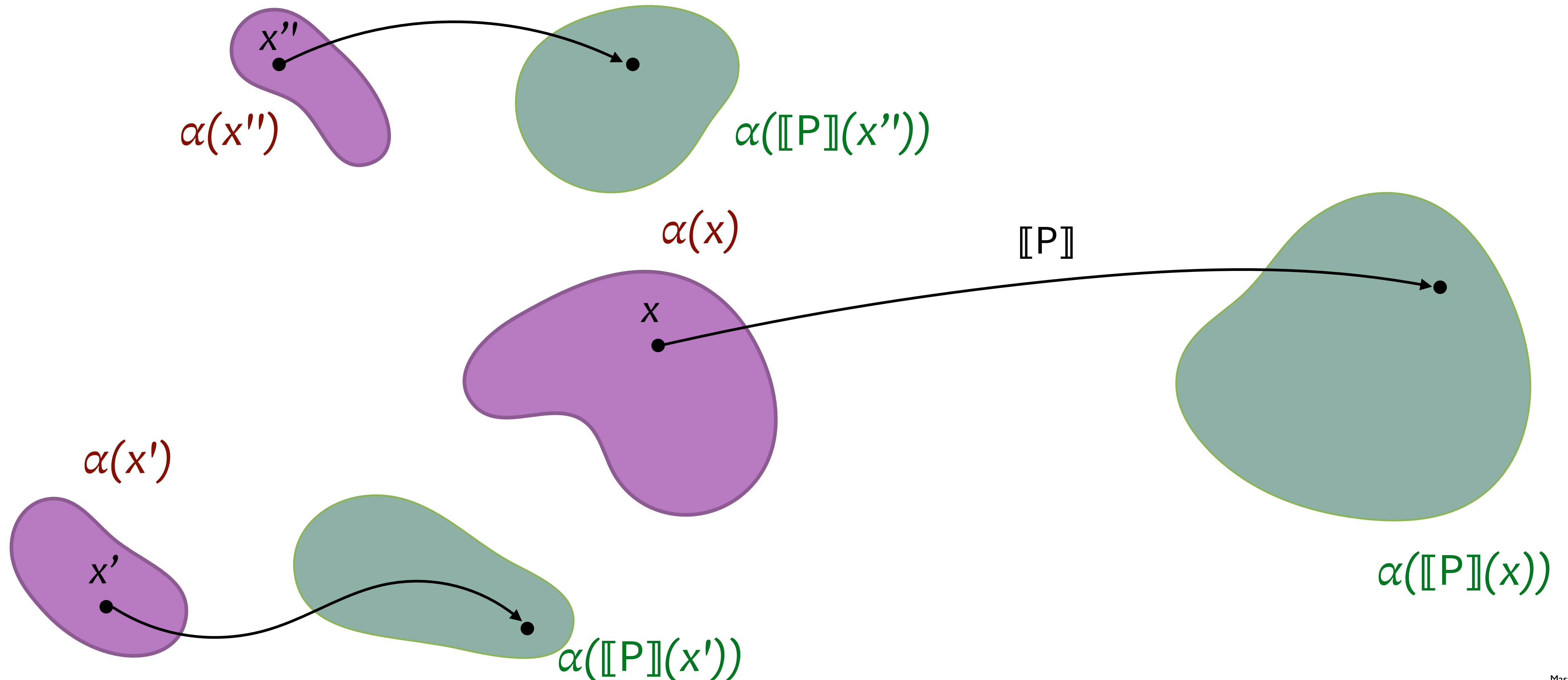


➤ Corresponds to Local completeness on all y such that $\alpha(y) = \alpha(x)$

COMPLETENESS AS HYPERPROPERTY

[GM18]

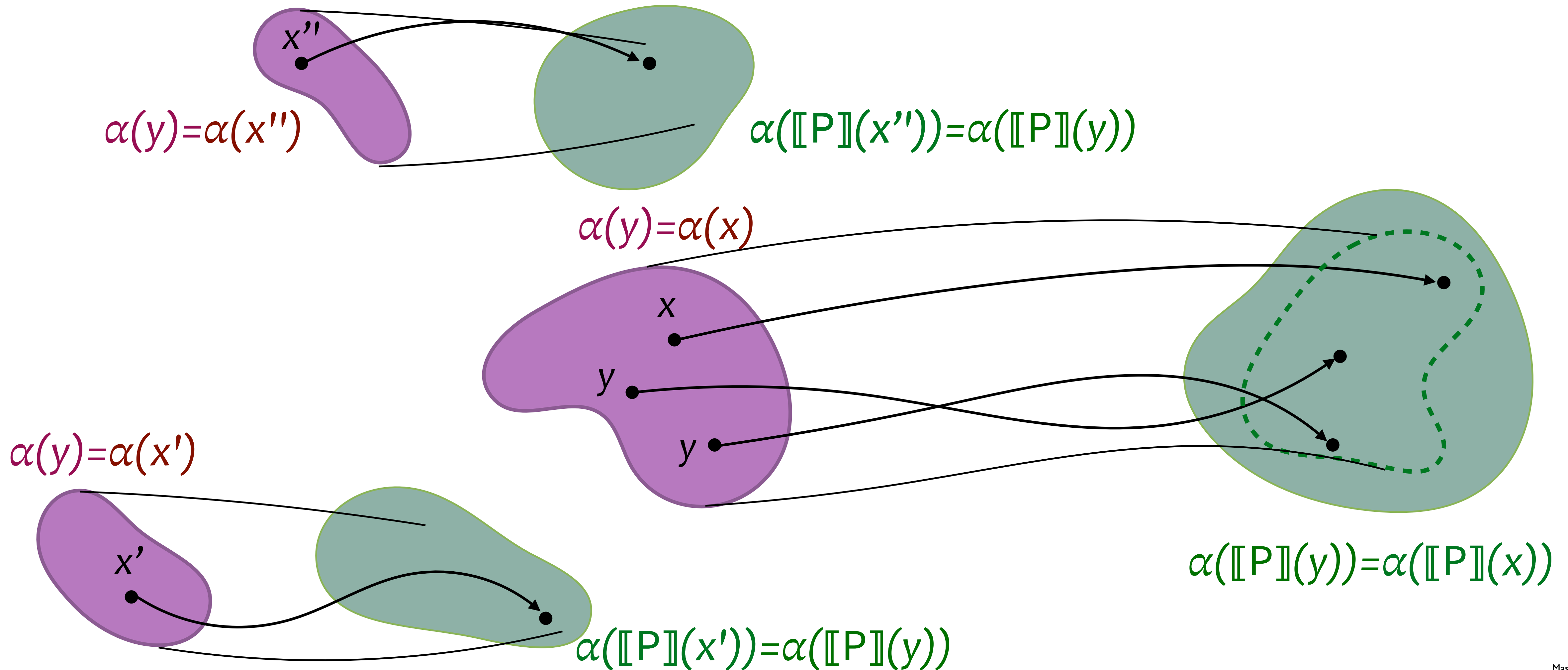
- Abstract Non Interference (ANI), $A = \gamma\alpha: \forall x$



COMPLETENESS AS HYPERPROPERTY

[GM18]

- Abstract Non Interference (ANI), $A = \gamma\alpha$: $\forall x$ and $\forall y \equiv \text{Completeness } (\forall x)$

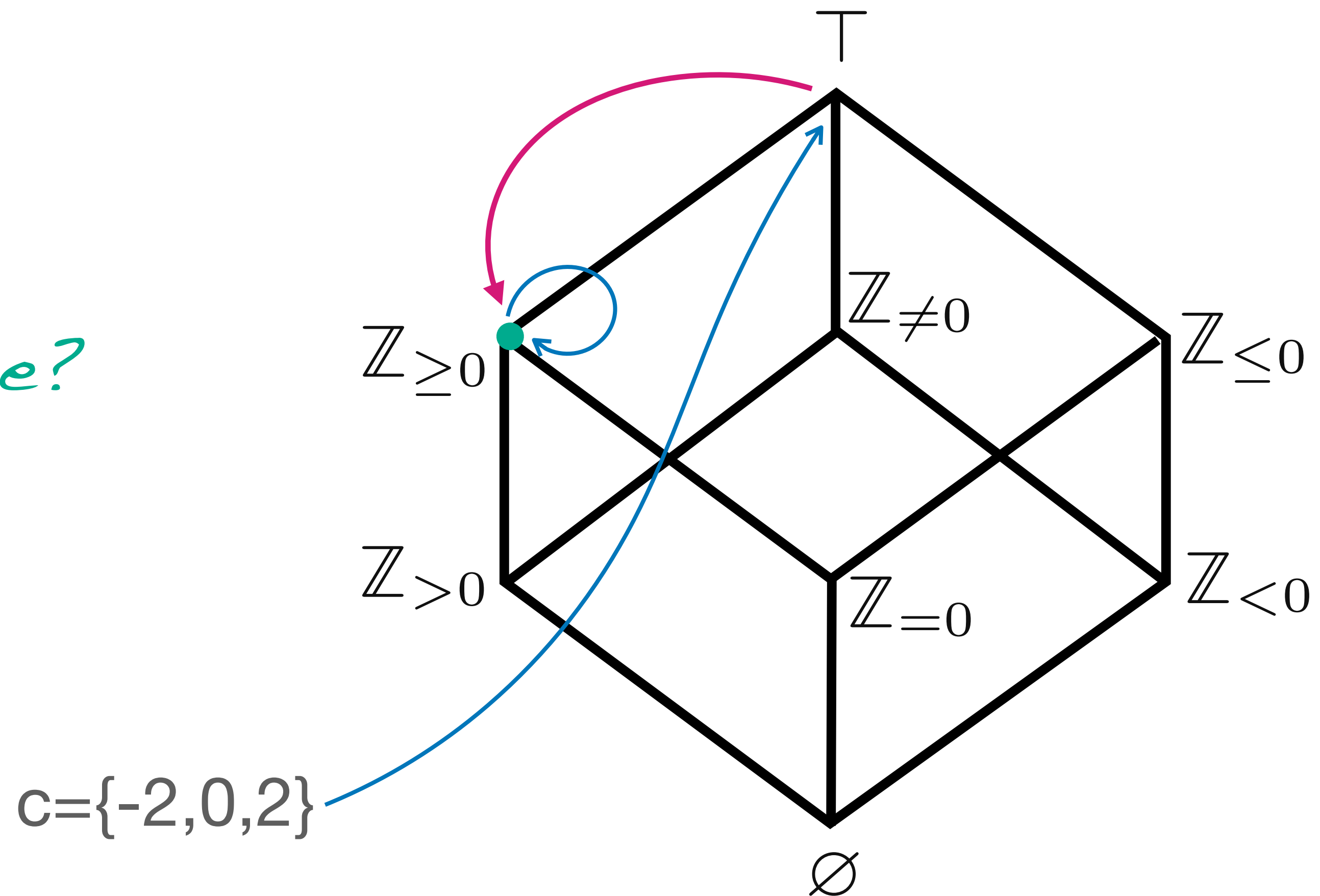


AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

(Global) Complete?

$\text{Sign} \circ f \circ \text{Sign}(c)?$



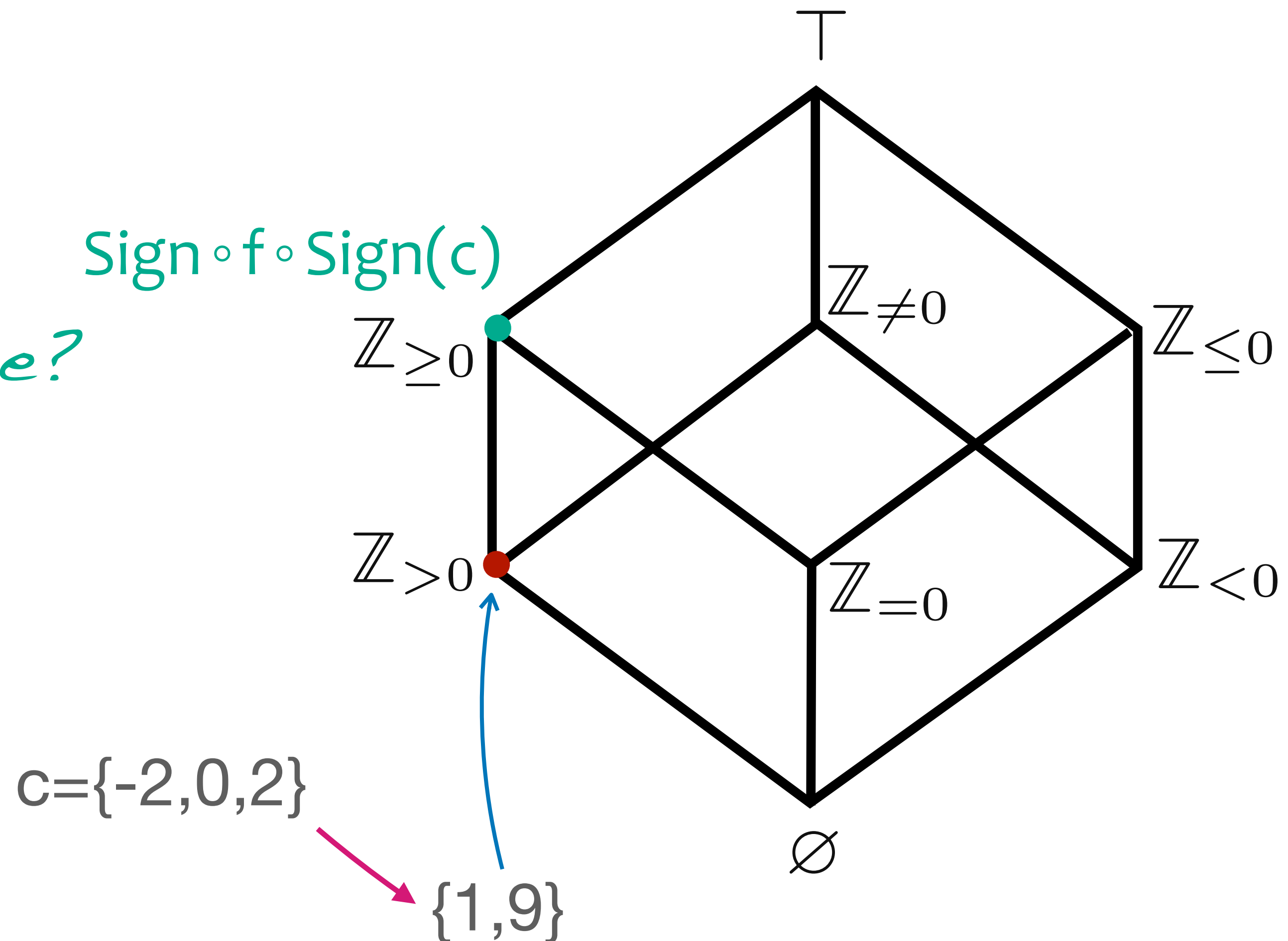
$c = \{-2, 0, 2\}$

AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

(Global) **No!** Complete?

$\text{Sign} \circ f(c)?$

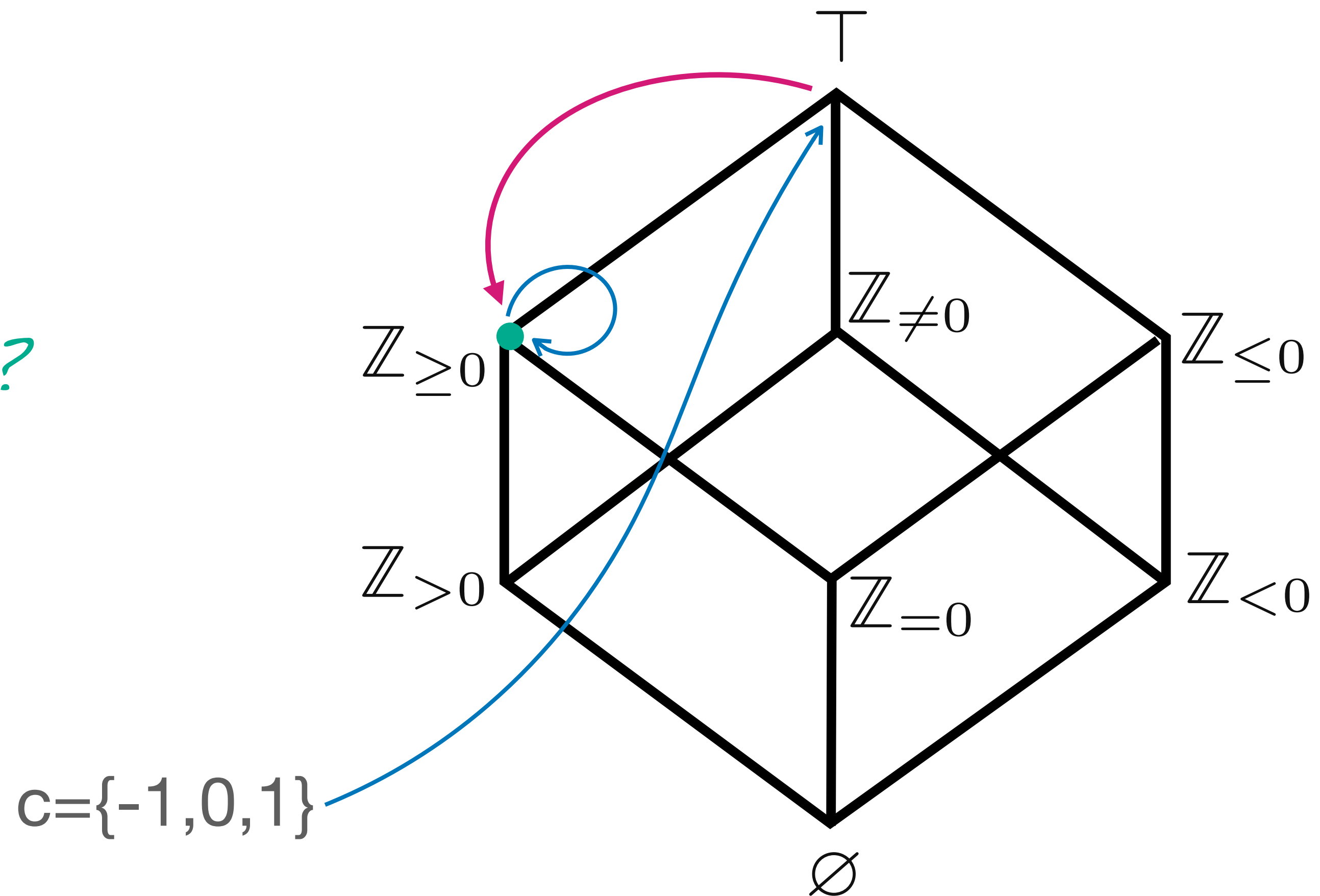


AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

Local Complete?

$\text{Sign} \circ f \circ \text{Sign}(c)?$

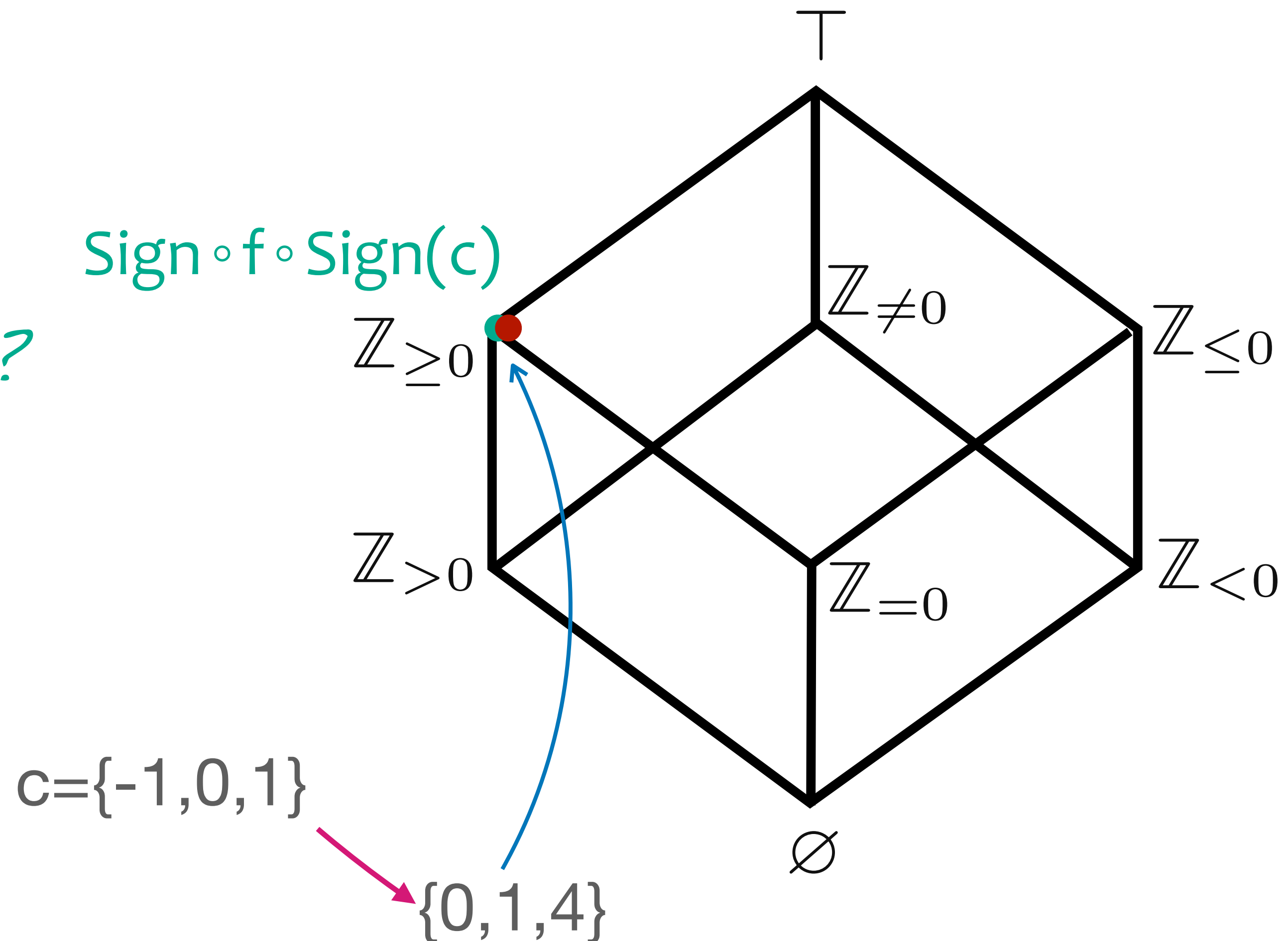


AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

*Local **Yes!** Complete?*

Sign $\circ f(c)$?



AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

*Local **Yes!** Complete?*

$$\text{Sign}(\{-1, 0, 1\}) = \text{Sign}(\{-2, 0, 2\})$$



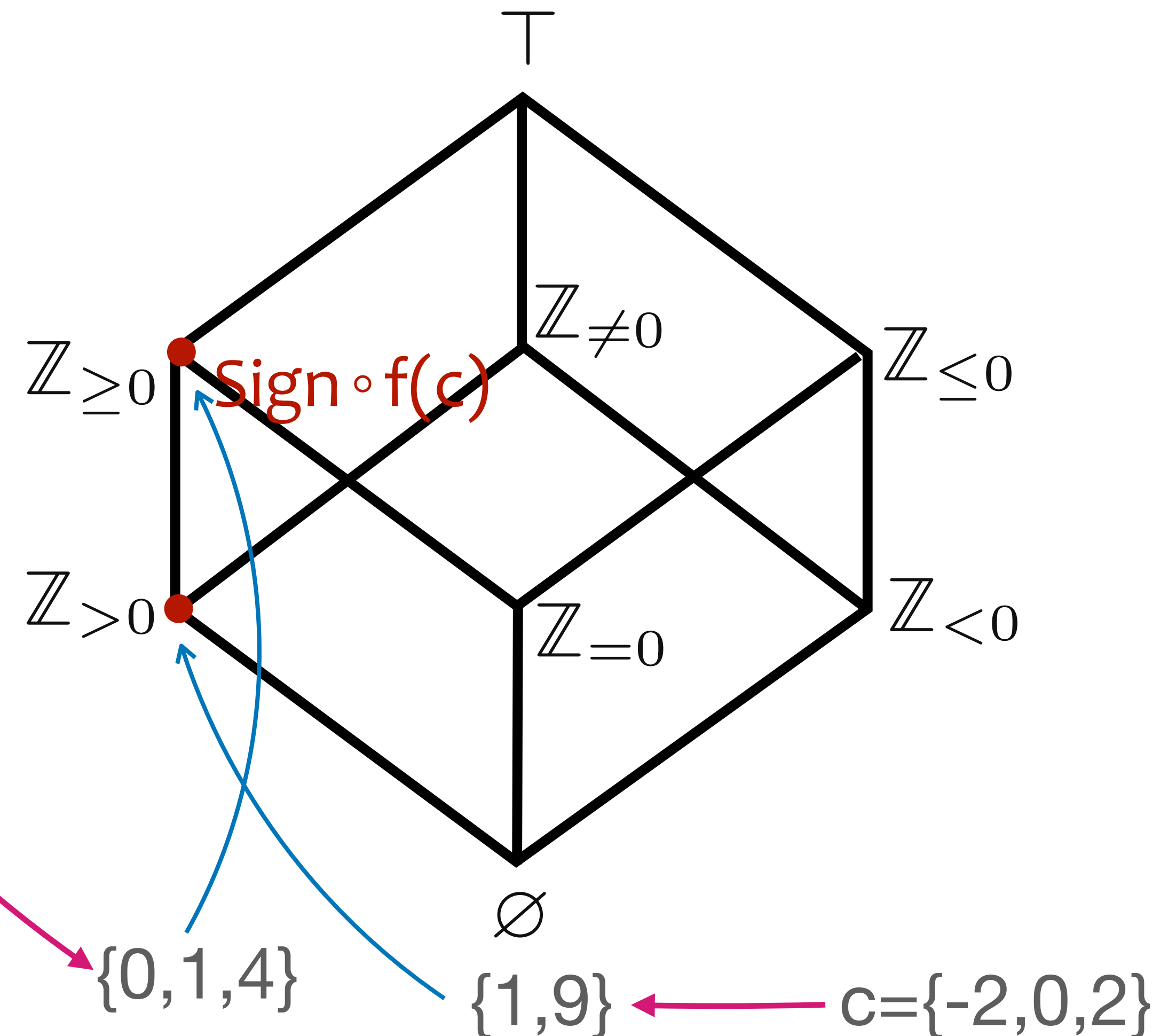
*Abstract Local **No!** Complete*

$$c = \{-1, 0, 1\}$$

$$\{0, 1, 4\}$$

$$\{1, 9\}$$

$$c = \{-2, 0, 2\}$$

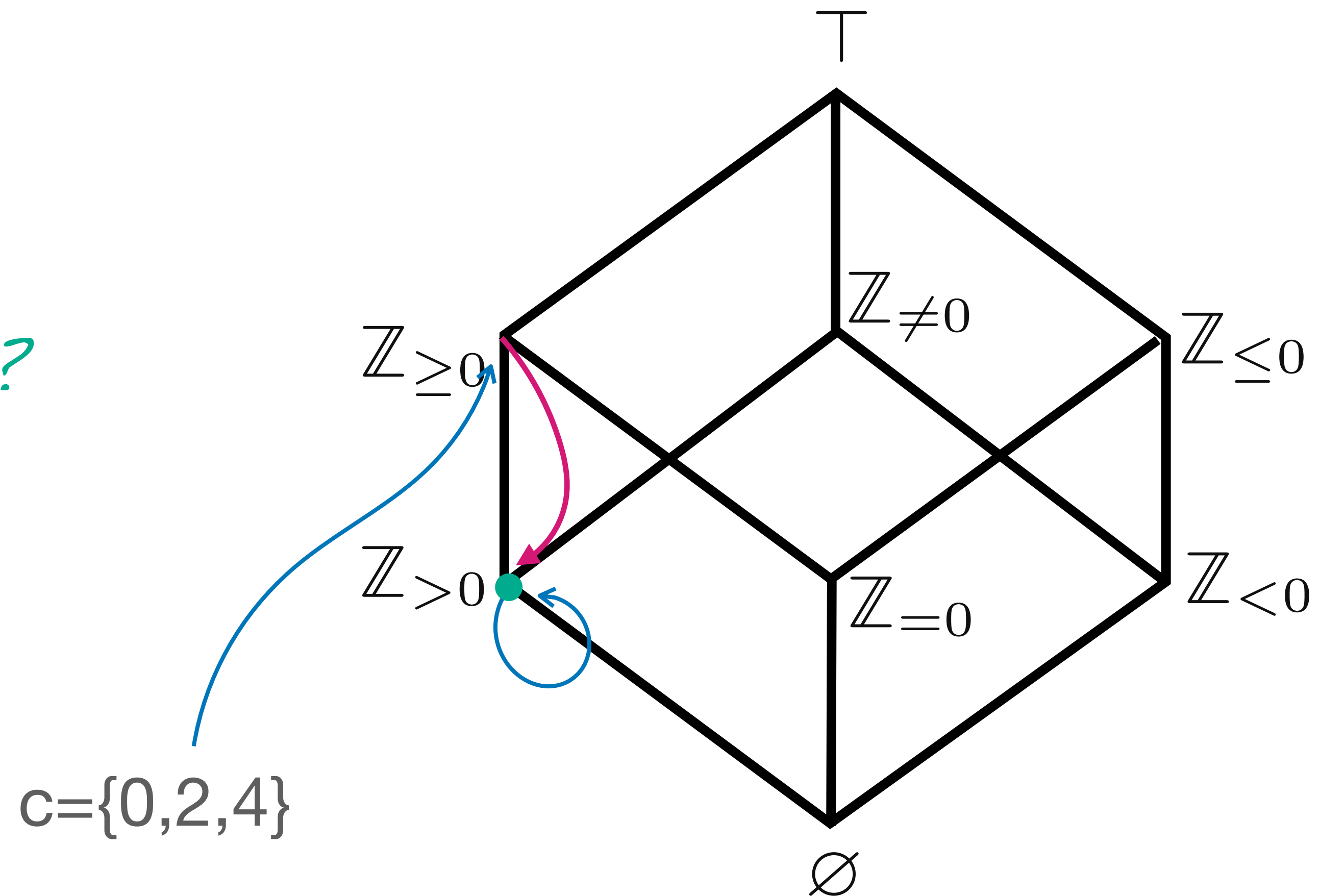


AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

Local Complete?

$\text{Sign} \circ f \circ \text{Sign}(c)?$



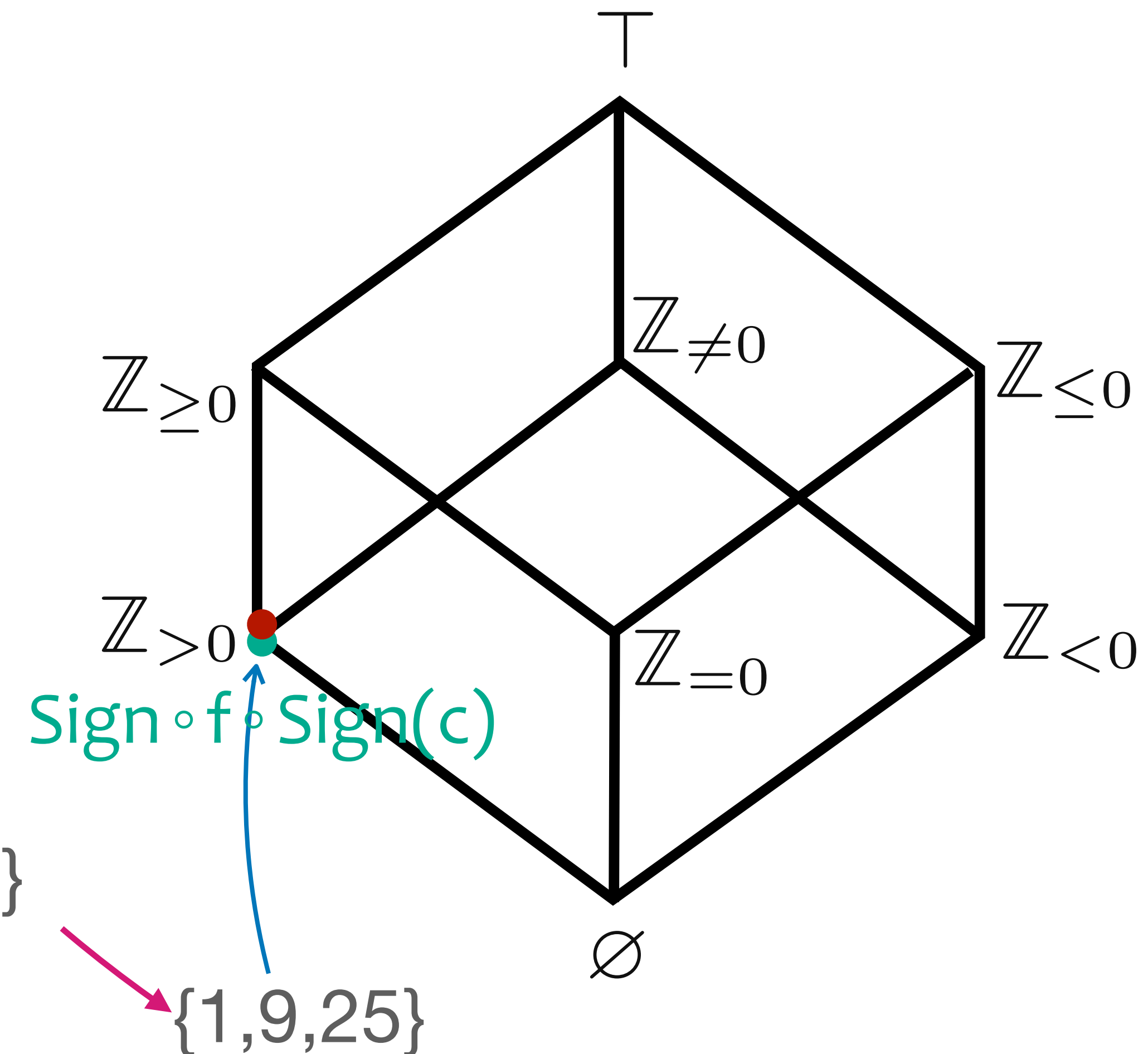
AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

Local Yes!

$\text{Sign} \circ f(c)?$

$c = \{0, 2, 4\}$



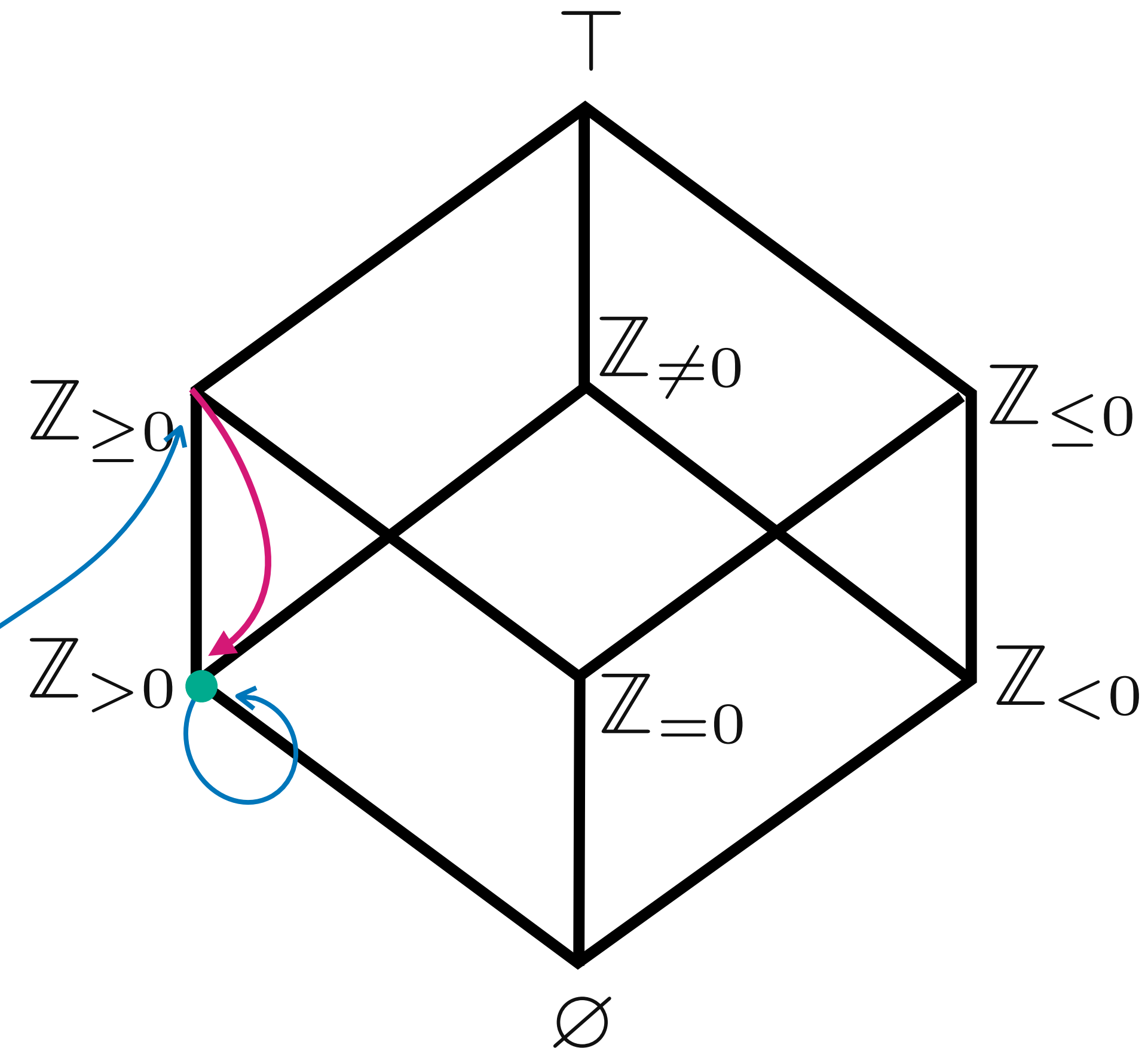
AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

Abstract Local Complete?

$\forall c'. \text{Sign}(c') = \text{Sign}(c). \text{Sign} \circ f \circ \text{Sign}(c')$?

$c' = \{0, 2, 3, 4\}$



AN EXAMPLE

- Consider the **Sign** domain and the function $f = \lambda c. \{(x+1)^2 \mid x \in c\}$

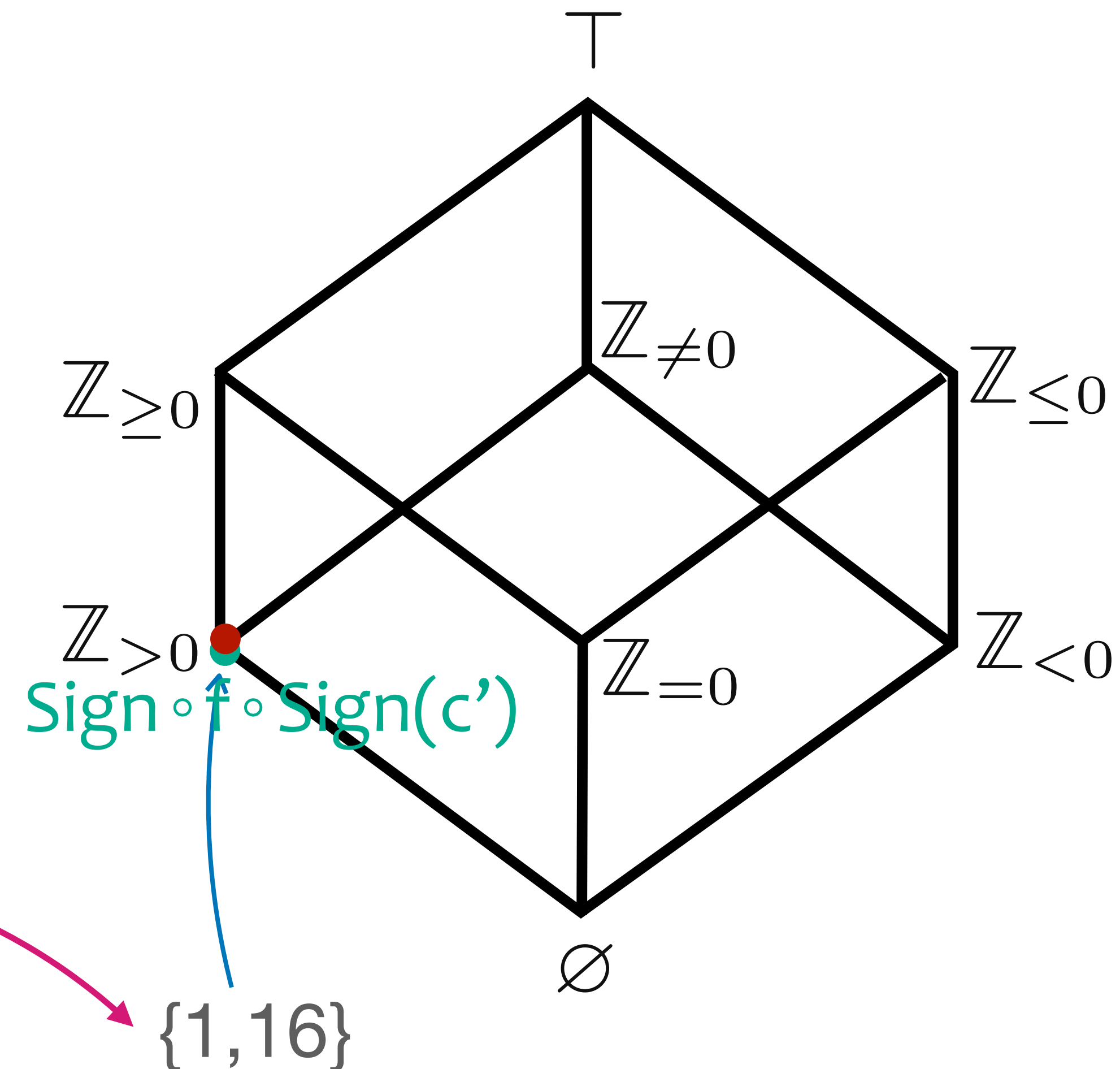
$$c = \{0, 2, 4\}$$

Abstract **Yes!** Complete?

$$\forall c'. \text{Sign}(c') = \text{Sign}(c). \text{Sign} \circ f(c')?$$

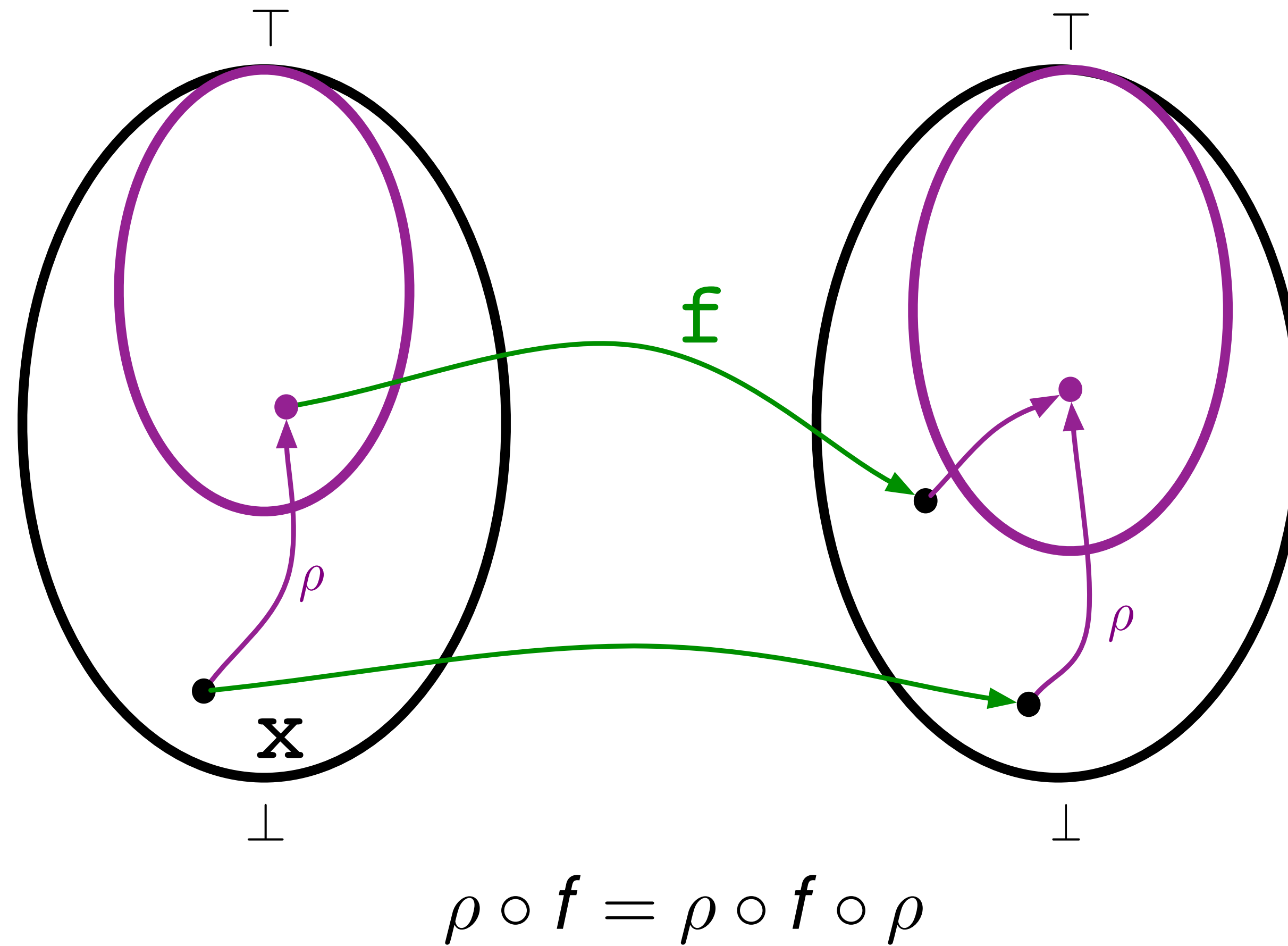
$$\text{Sign} \circ f \circ \text{Sign}(c') = \text{Sign} \circ f(c')$$

$$c' = \{0, 3\}$$



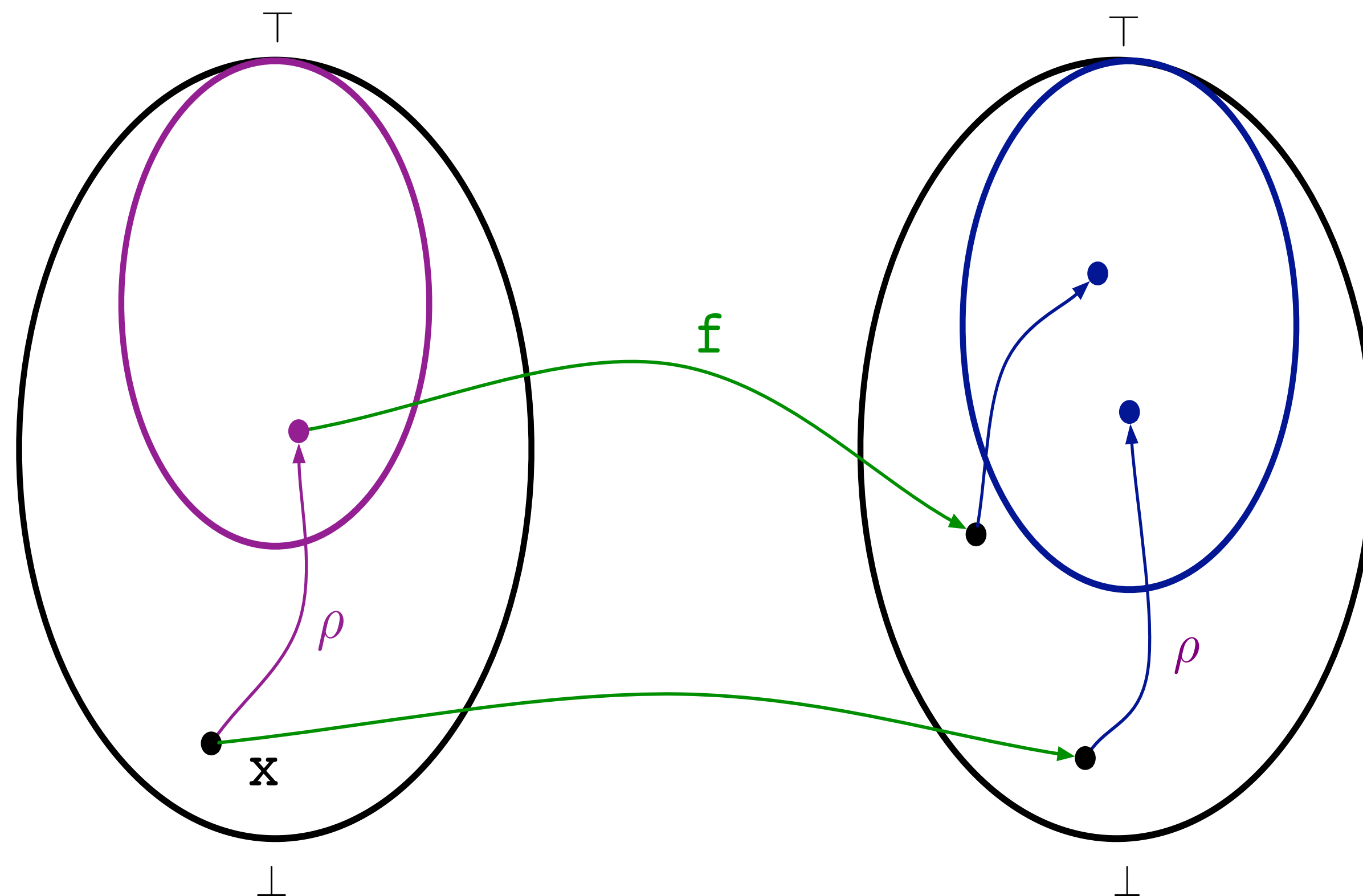
MAKING DOMAINS (BACKWARD) COMPLETE

IDEA



MAKING DOMAINS (BACKWARD) COMPLETE

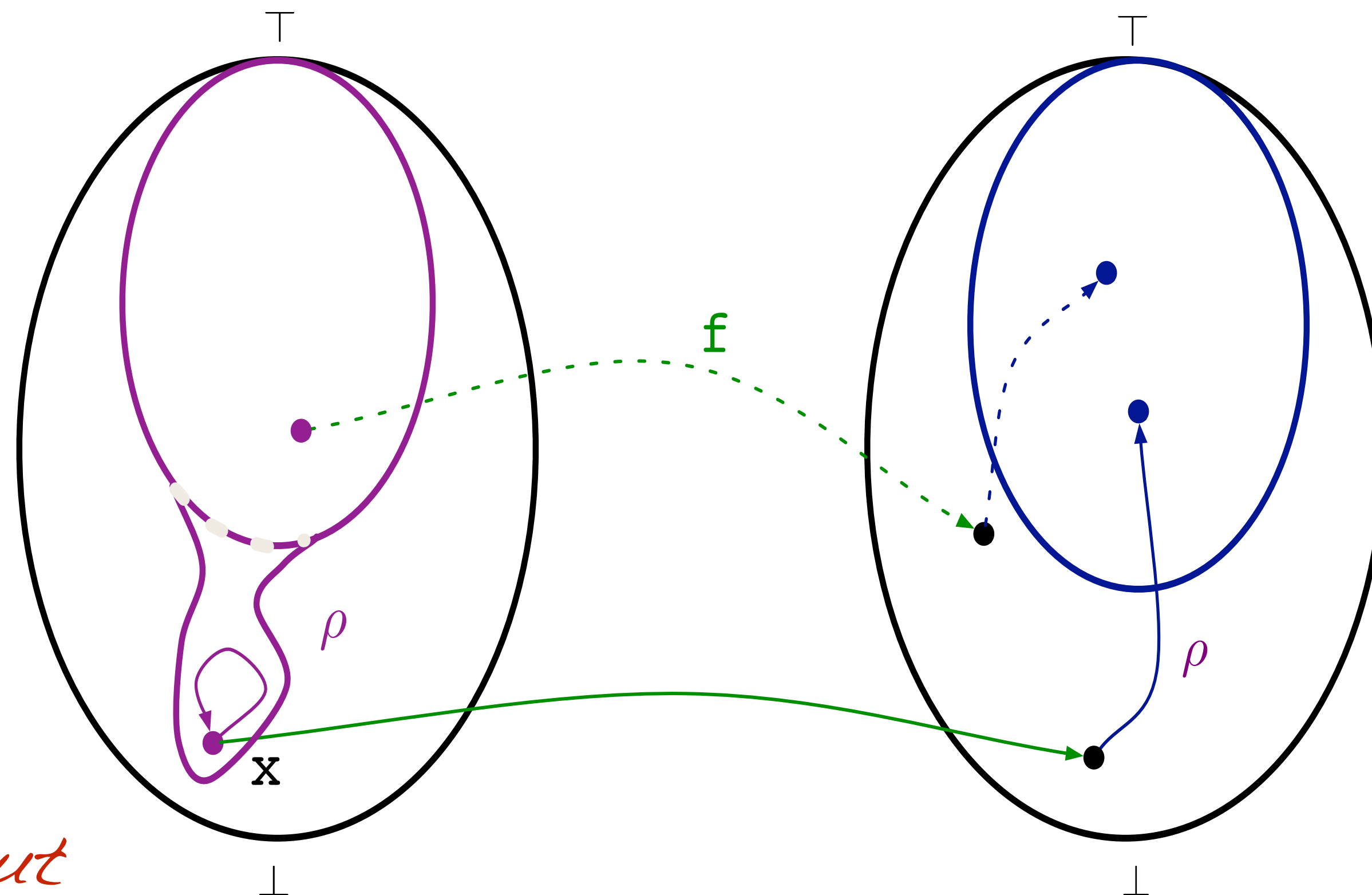
IDEA



$$\rho \circ f \neq \rho \circ f \circ \rho$$

MAKING DOMAINS (BACKWARD) COMPLETE

IDEA

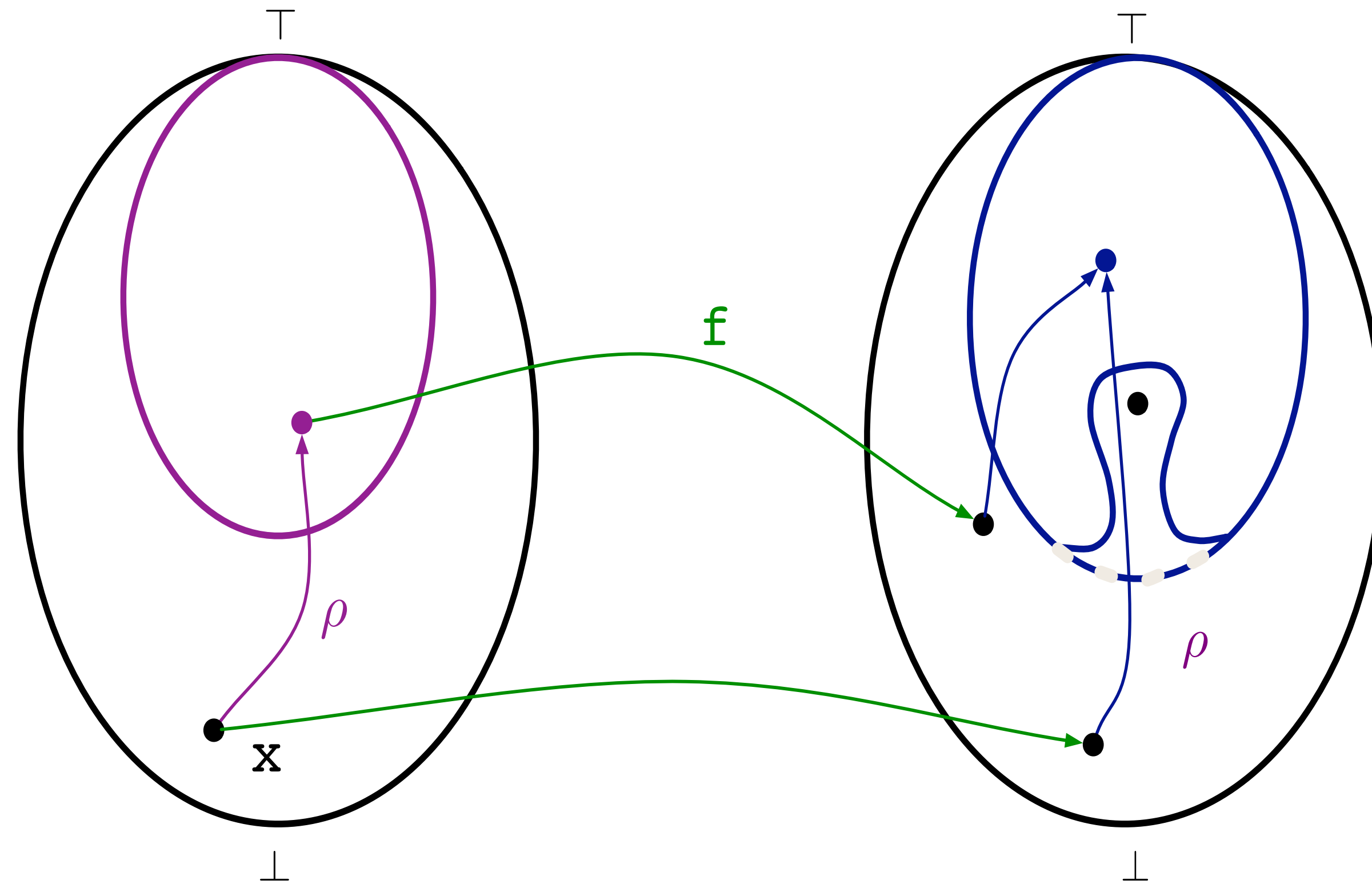


Refining the input domain

$$\rho \circ f = \rho \circ f \circ \rho$$

MAKING DOMAINS (BACKWARD) COMPLETE

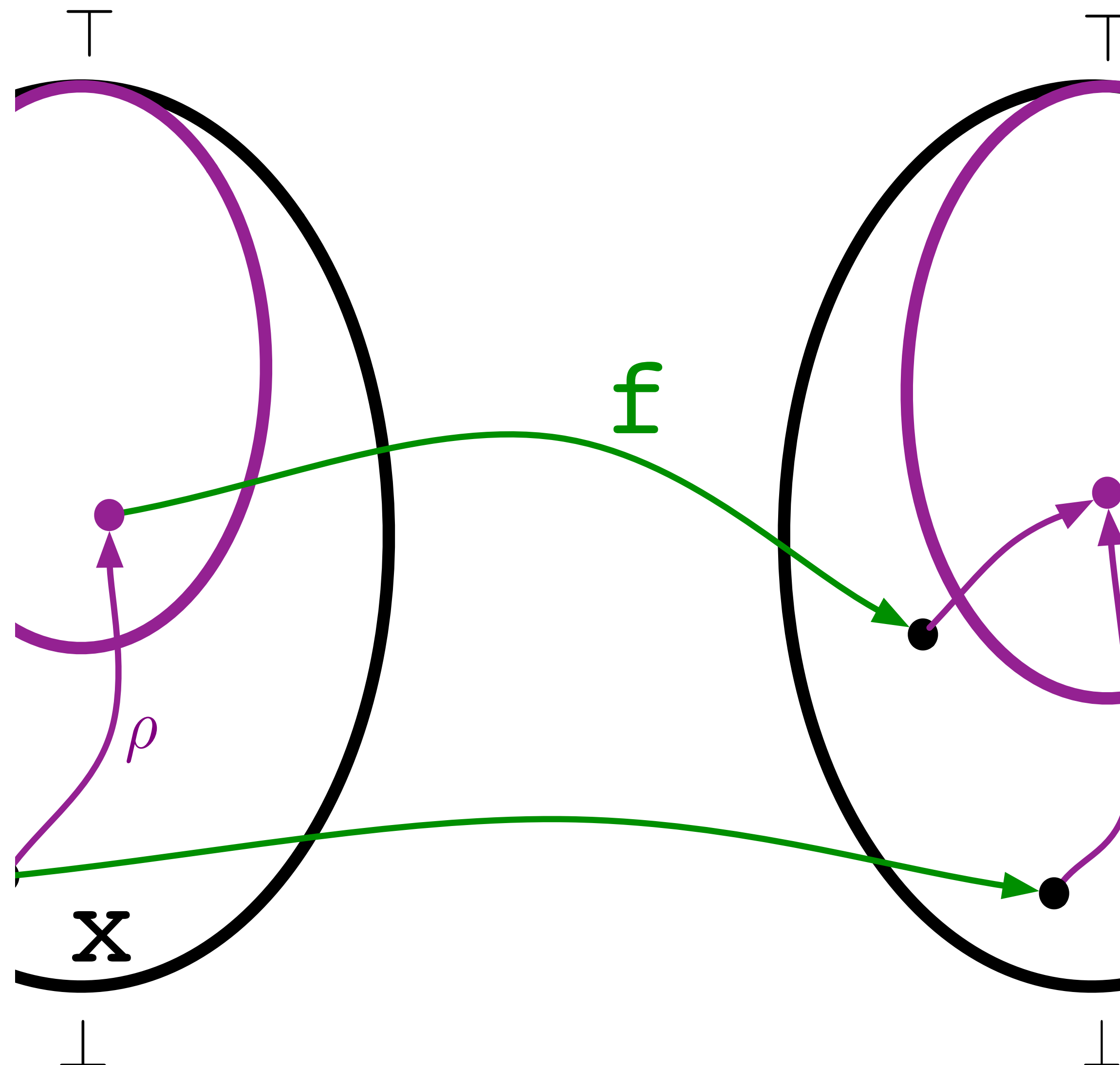
IDEA



$$\rho \circ f = \rho \circ f \circ \rho$$

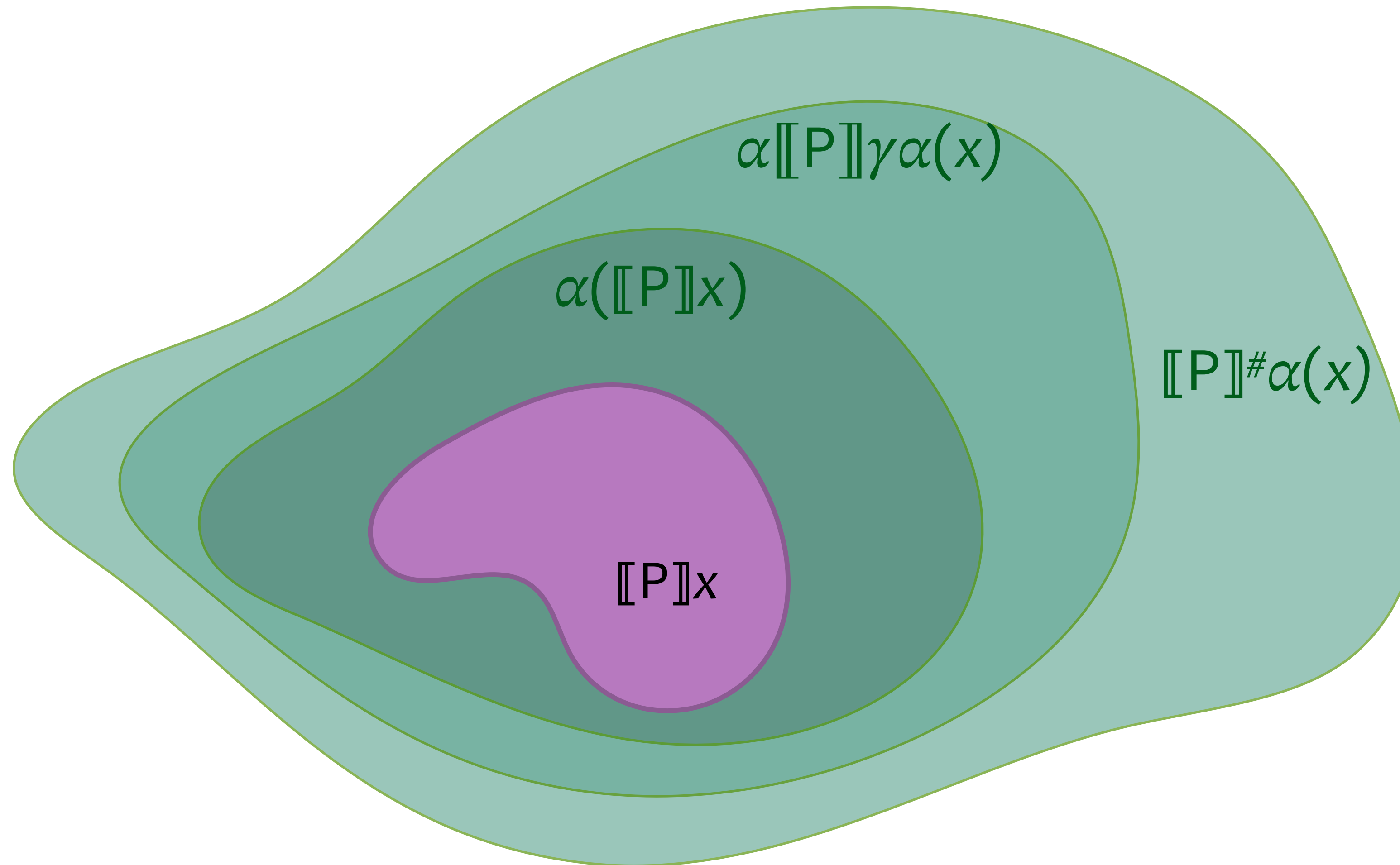
*Simplifying the
output domain*

COMPARING precisions



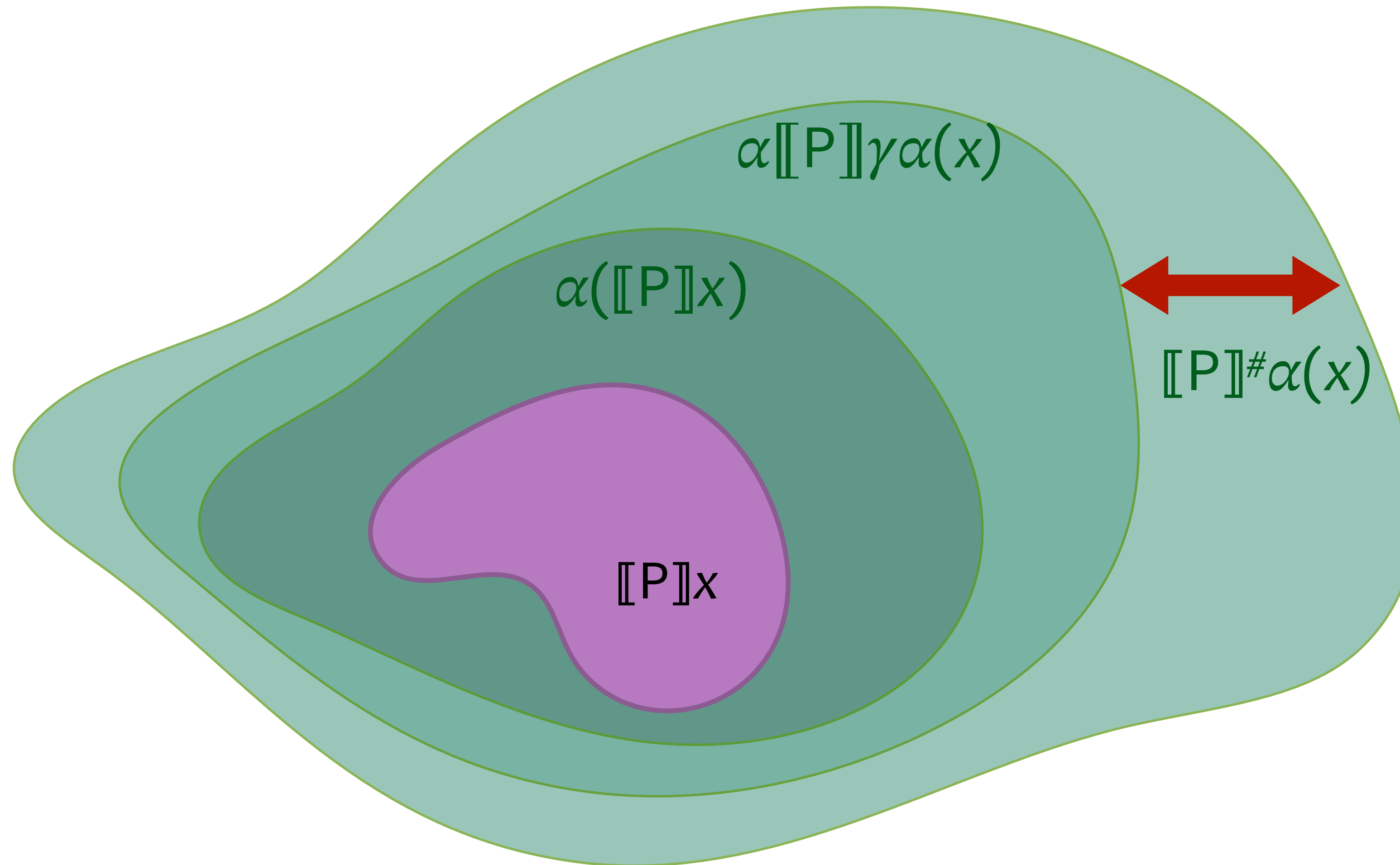
ABSTRACT COMPUTATIONS

[GM23]



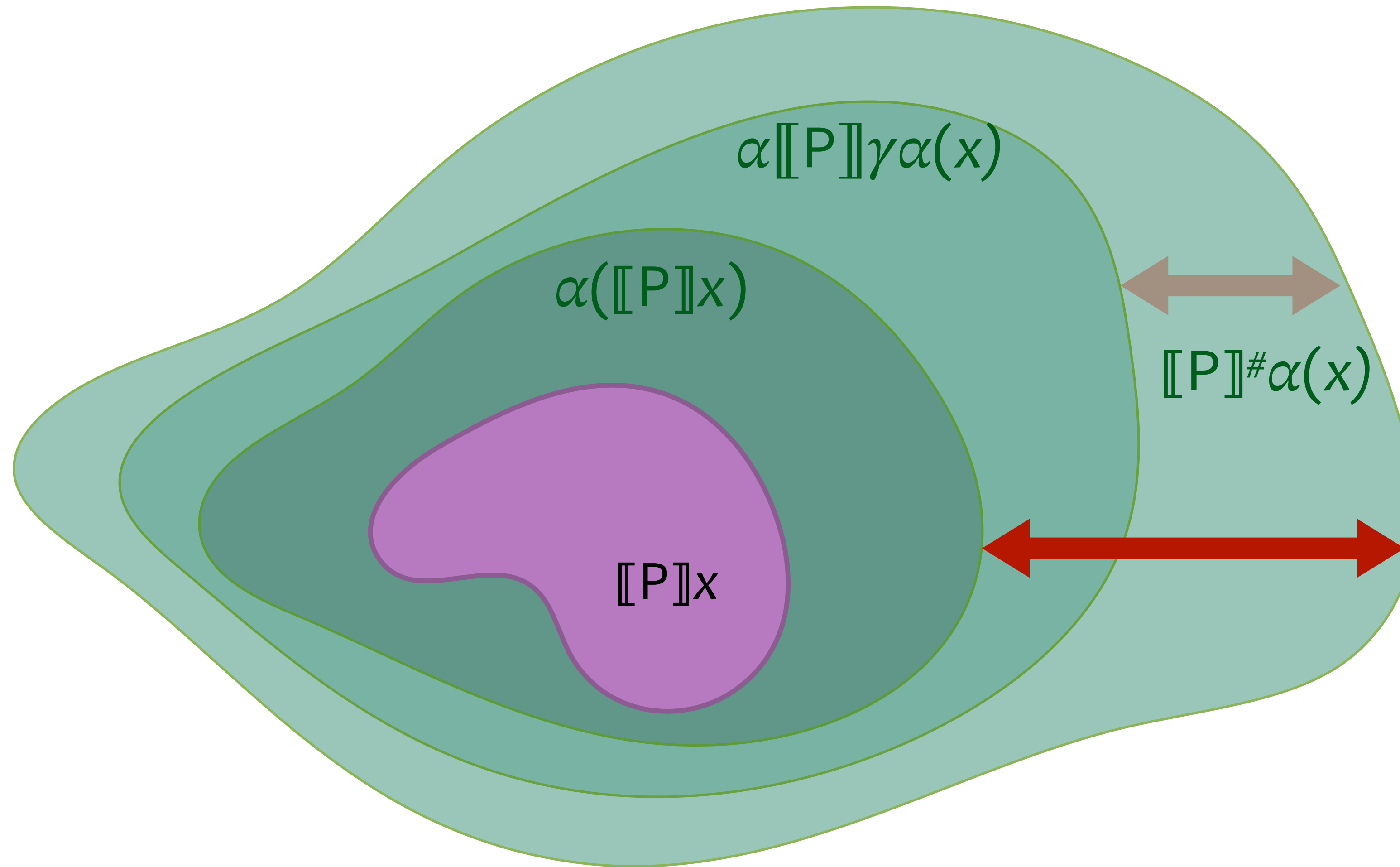
BCA PROPERTY

[GM23]



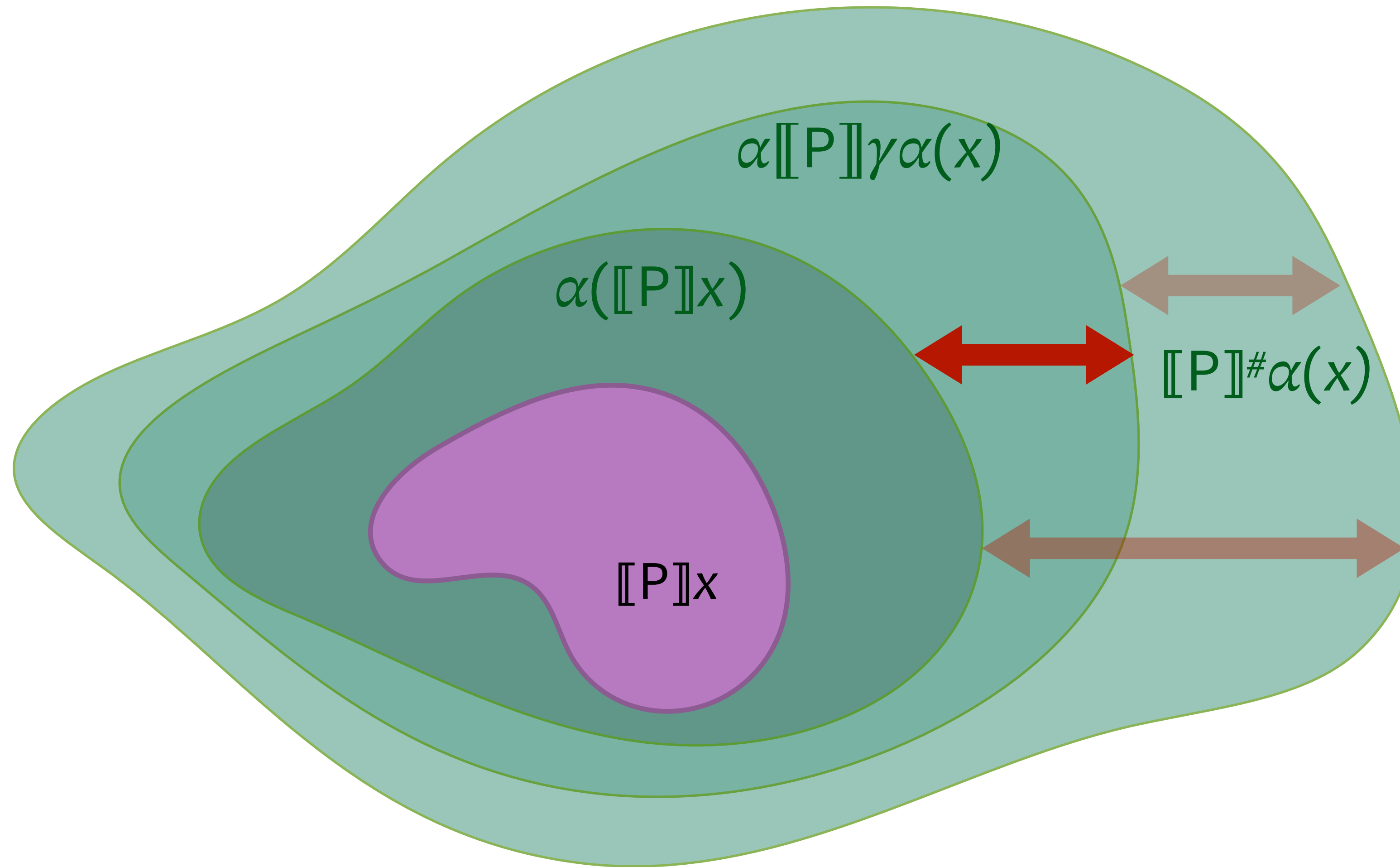
INCOMPLETENESS OF $\llbracket P \rrbracket^\#$

[GM23]



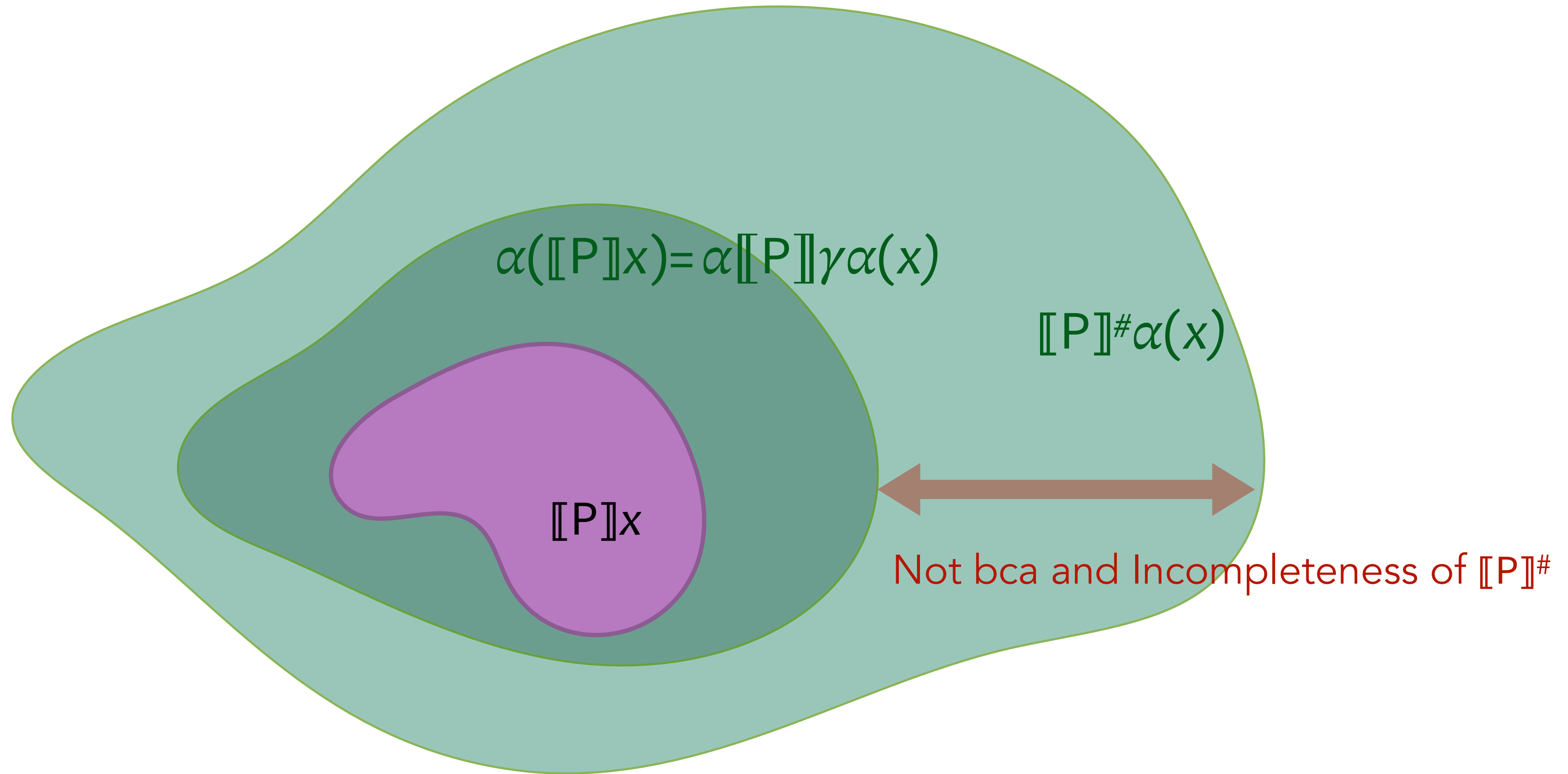
INCOMPLETENESS OF $\alpha\llbracket P \rrbracket\gamma$

[GM23]



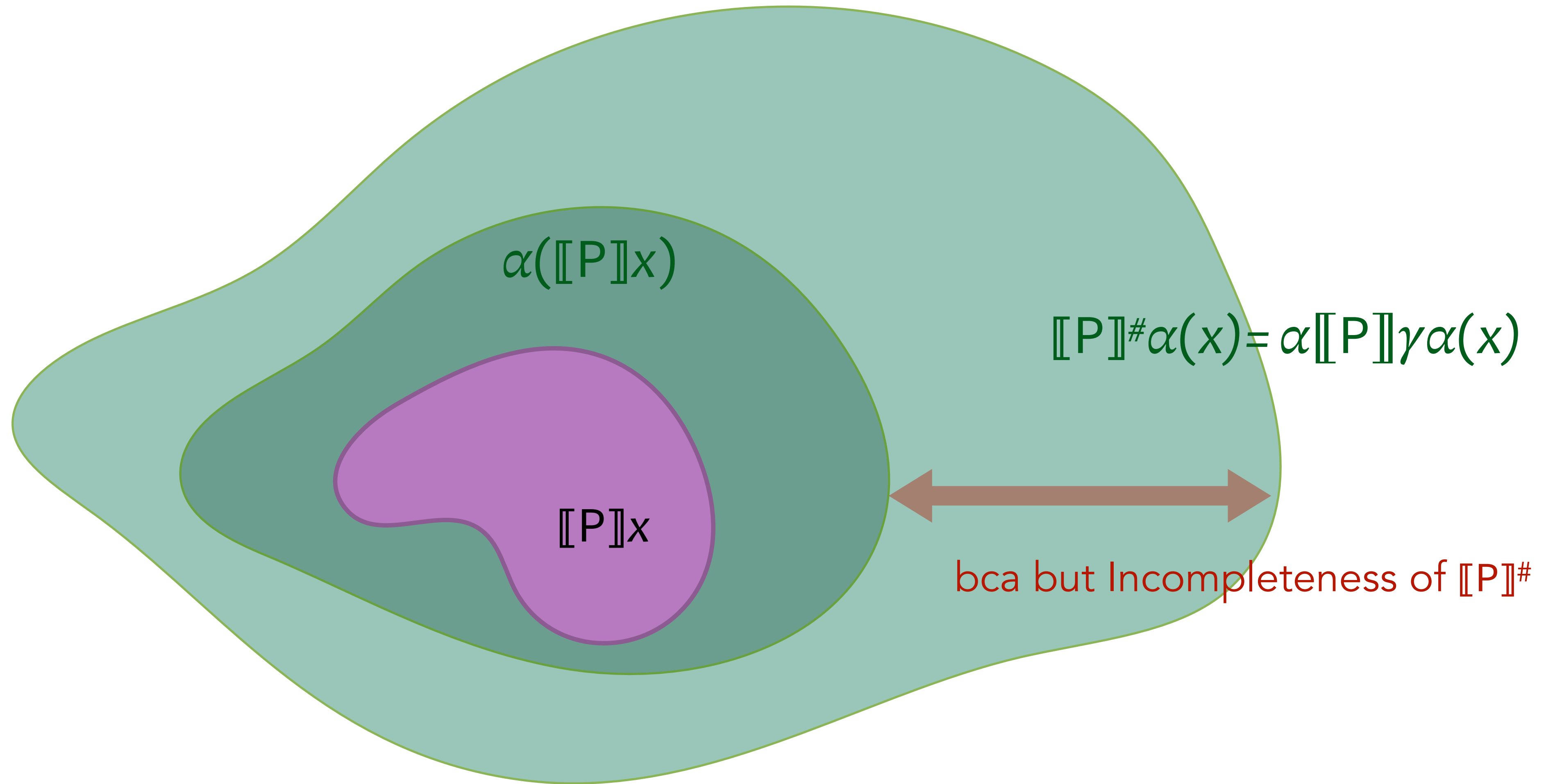
COMPLETENESS OF $\alpha[[P]]\gamma$

[GM23]



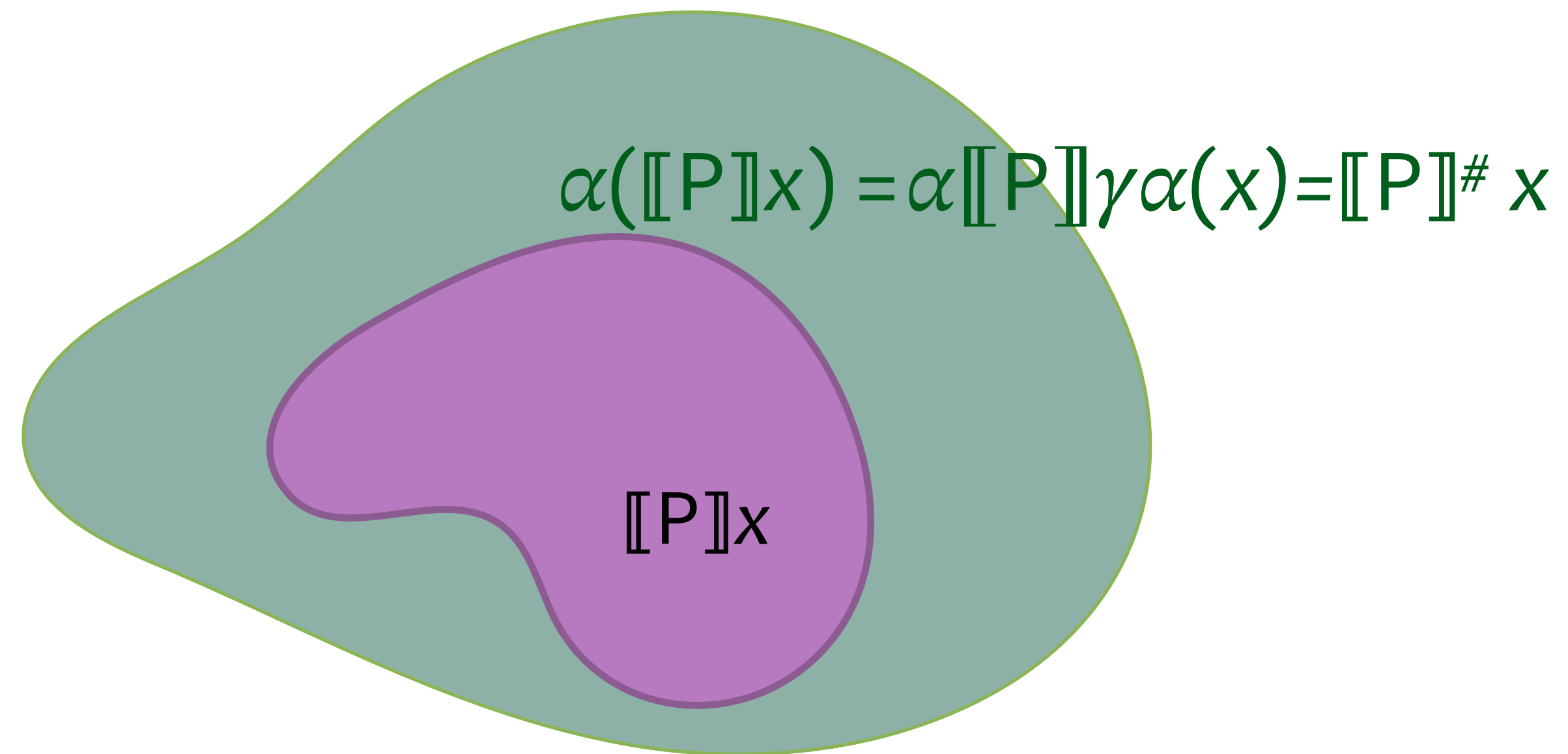
INCOMPLETENESS OF $\alpha\llbracket P \rrbracket\gamma$

[GM23]



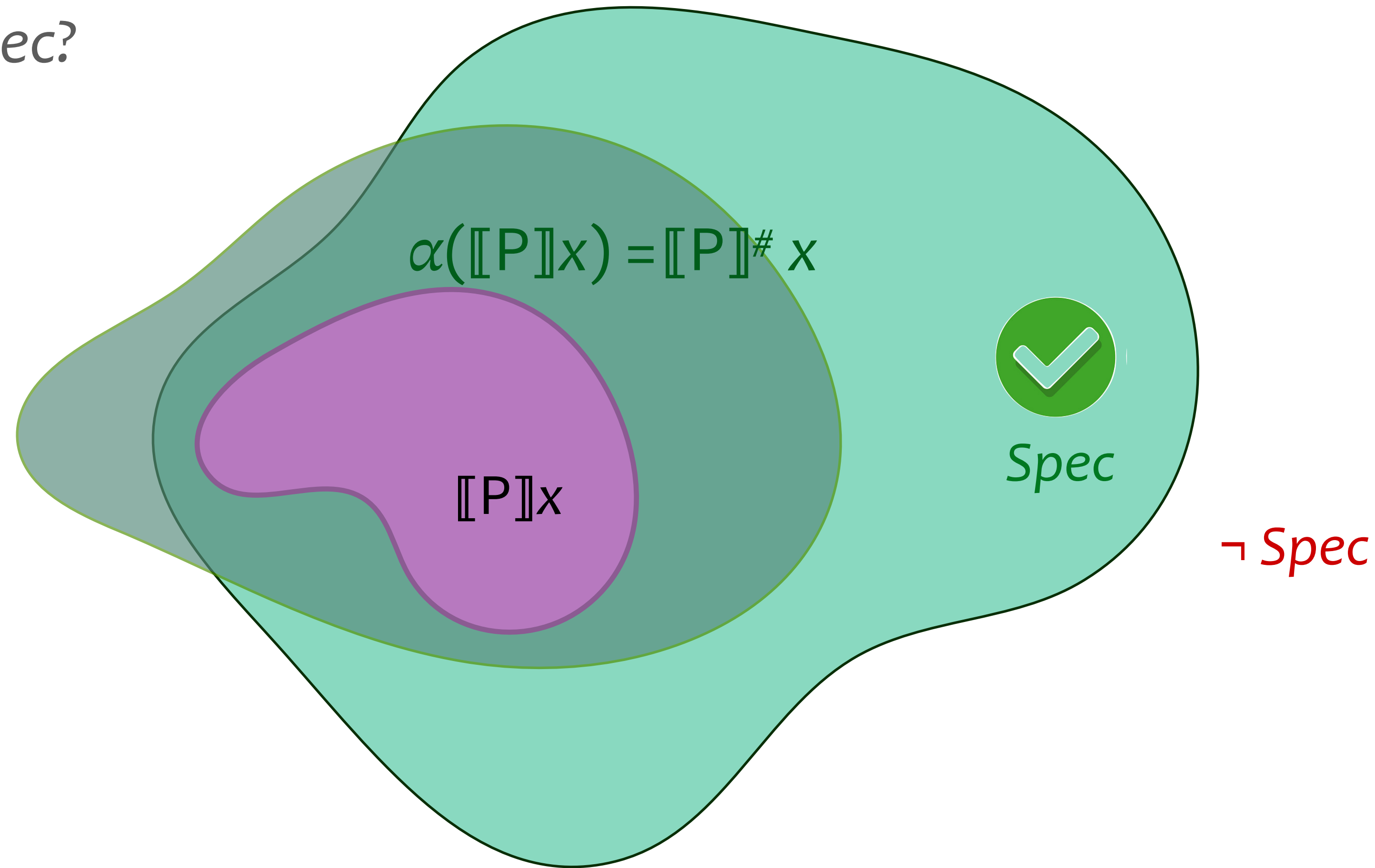
COMPLETENESS (AND BCA)

[GM23]

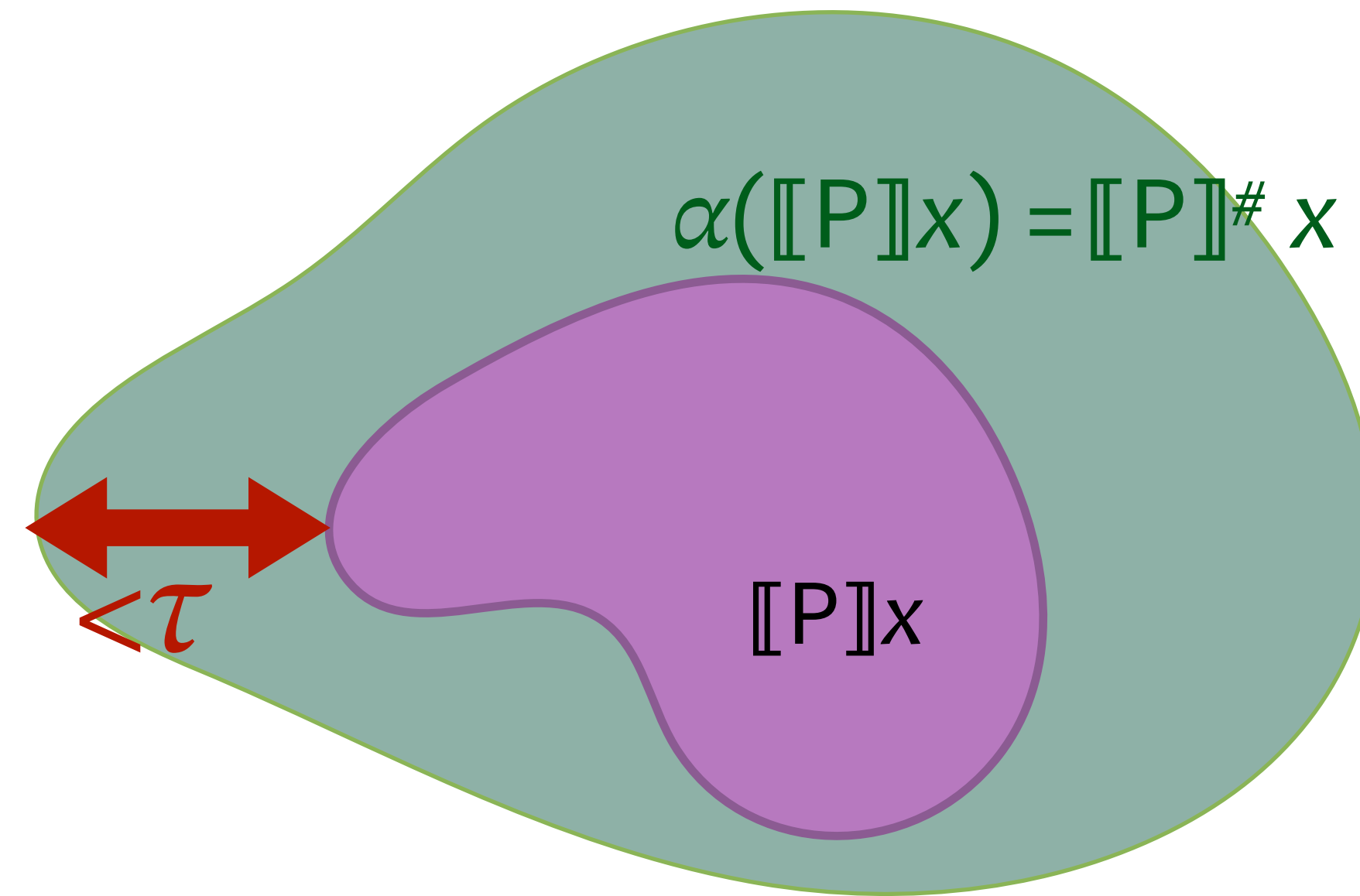


COMPLETENESS MAY BE NOT ENOUGH [GM23]

► $\llbracket P \rrbracket x \subseteq \text{Spec}$?



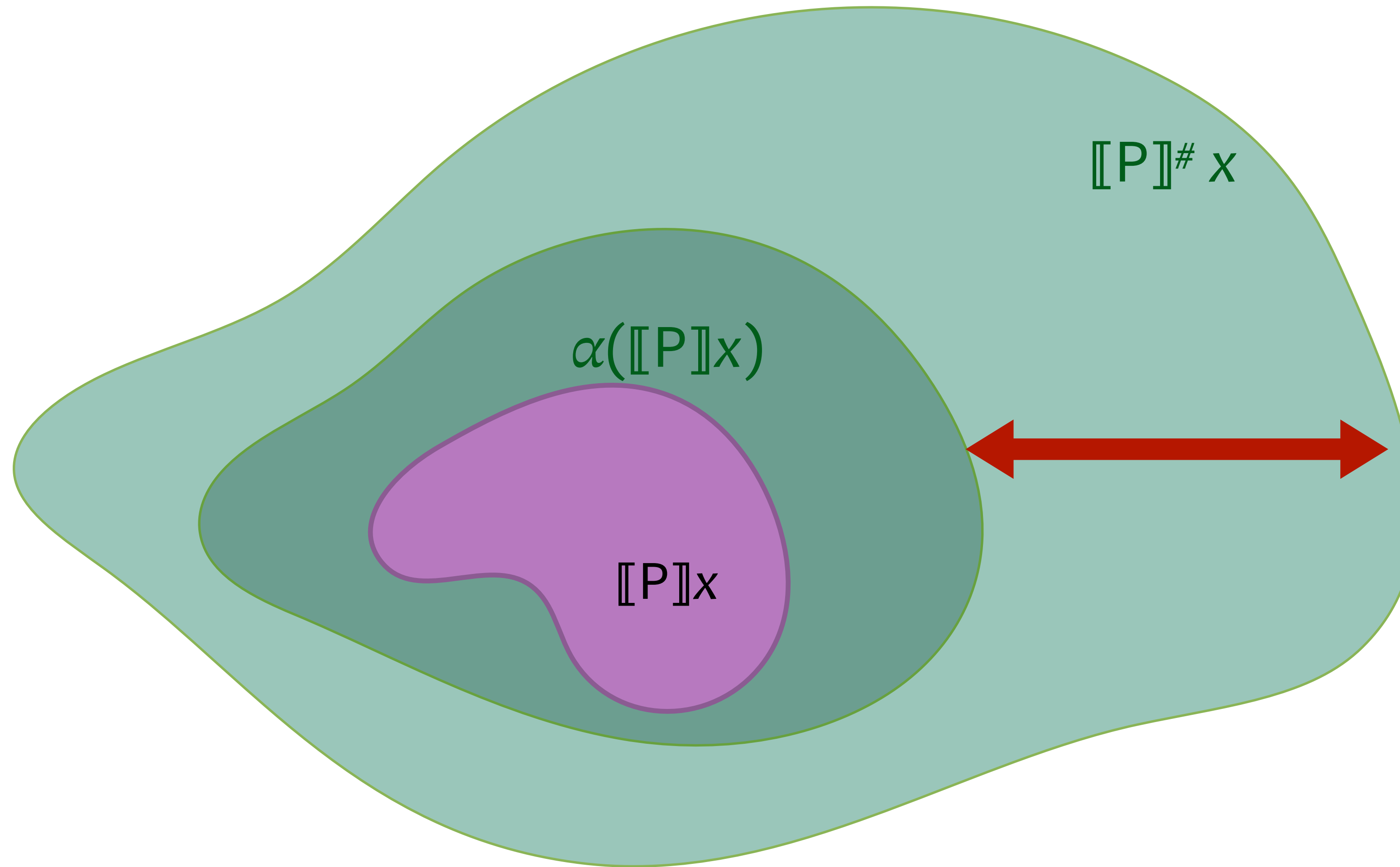
$\llbracket P \rrbracket x \subseteq \alpha(\llbracket P \rrbracket x) \not\subseteq \text{Spec}$



$$(\mathbf{Global}) \quad \tilde{\mathbb{C}}^A(f)_\tau \Leftrightarrow \forall \mathbf{c} \in \mathbf{C}. A \circ f \circ A(\mathbf{c}) \leq_{\mathbf{c}} \tau \text{ and}$$

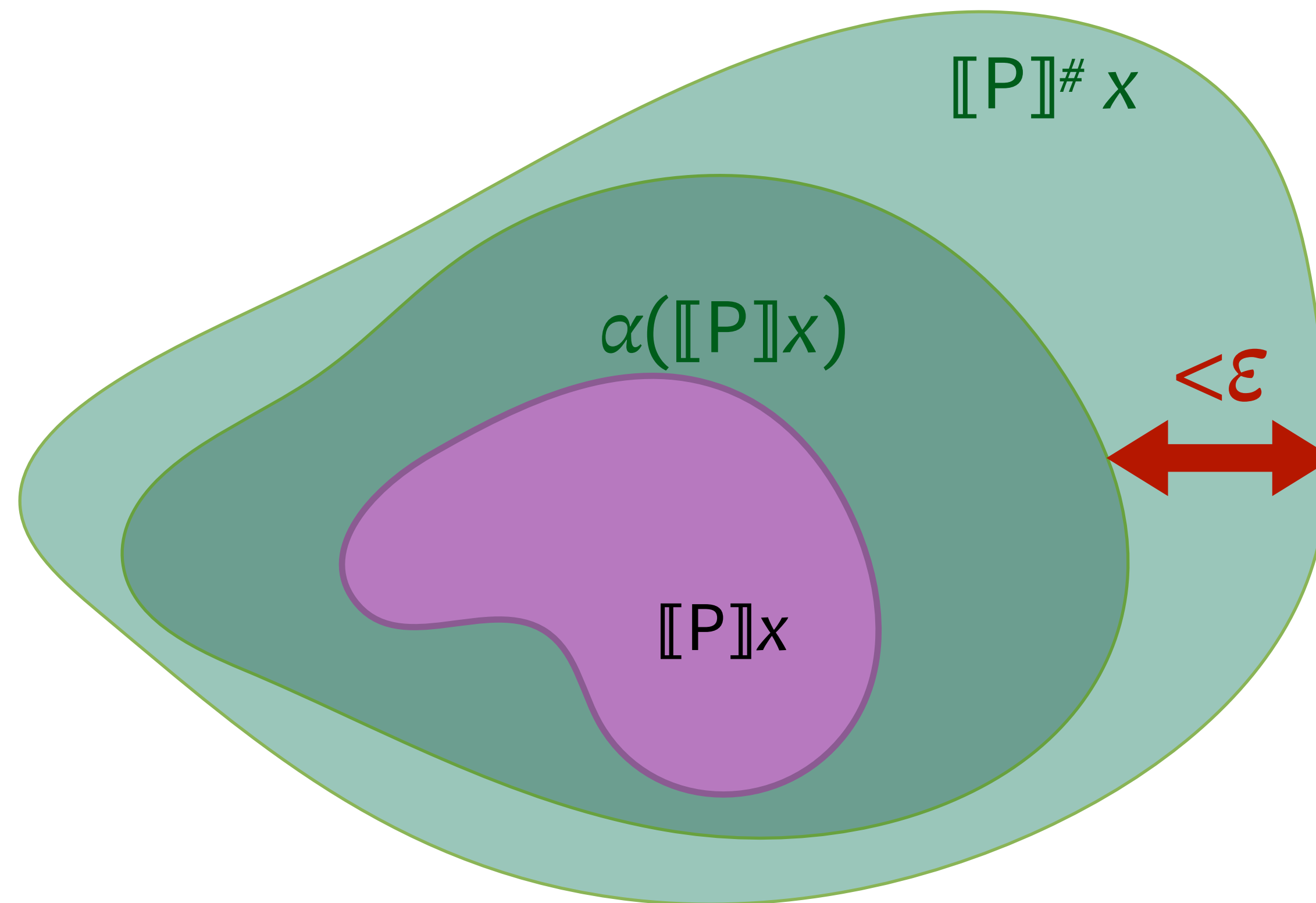
$$(\mathbf{Local}) \quad \tilde{\mathbb{C}}^A_c(f)_\tau \Leftrightarrow A \circ f \circ A(\mathbf{c}) \leq_{\mathbf{c}} \tau$$

INCOMPLETENESS OF MAY BE ACCEPTABLE



PARTIAL COMPLETENESS OF $\llbracket P \rrbracket^\#$

[CDG22]



$$\delta(\alpha(\llbracket P \rrbracket x), \llbracket P \rrbracket^\# x) \leq \varepsilon$$

THANKS!